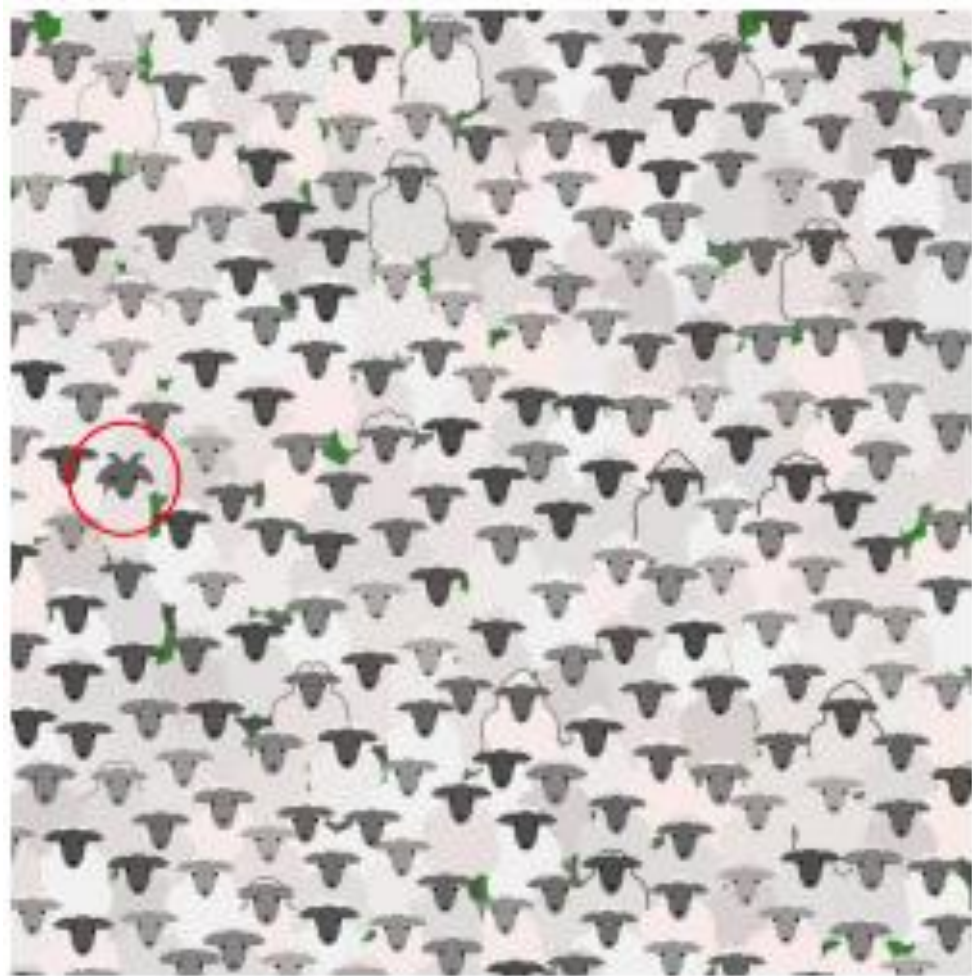


Anomaly (Novelty) Detection I

박성호

컴퓨터공학부



Agenda

1. Anomaly Detection
2. Density Estimation-based approaches
3. Distance-based approaches
4. Decision Boundary-based approaches
5. Reconstruction-based Approaches
6. 관련 연구사례 소개

Anomaly (Novelty) Detection?

- What is Abnormal/Novel Data (Outlier)?

“Observations that deviate so much from other observations as to arouse suspicions that they were generated by a different mechanism (Hawkins, 1980)”

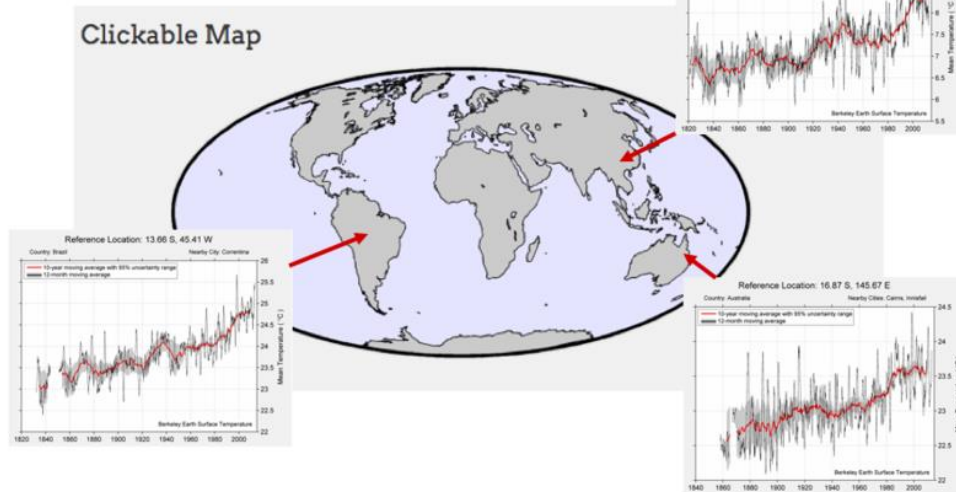
“Instances that their true probability density is very low (Harmeling et al., 2006)”

- Abnormal Data/Novel Data/Outliers are different from noise data
 - ✓ Noise is random error or variance in a measured variable
- Abnormal Data/Novel Data/Outliers are interesting
 - ✓ It violates the mechanism that generates the normal data

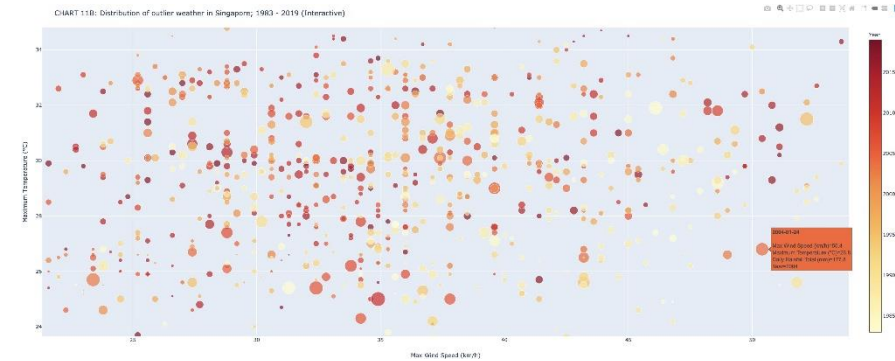
Anomaly (Novelty) Detection 사례

- 이상탐지 사례: 기후 이상

Weather novelty detection



Detecting outlier weather patterns

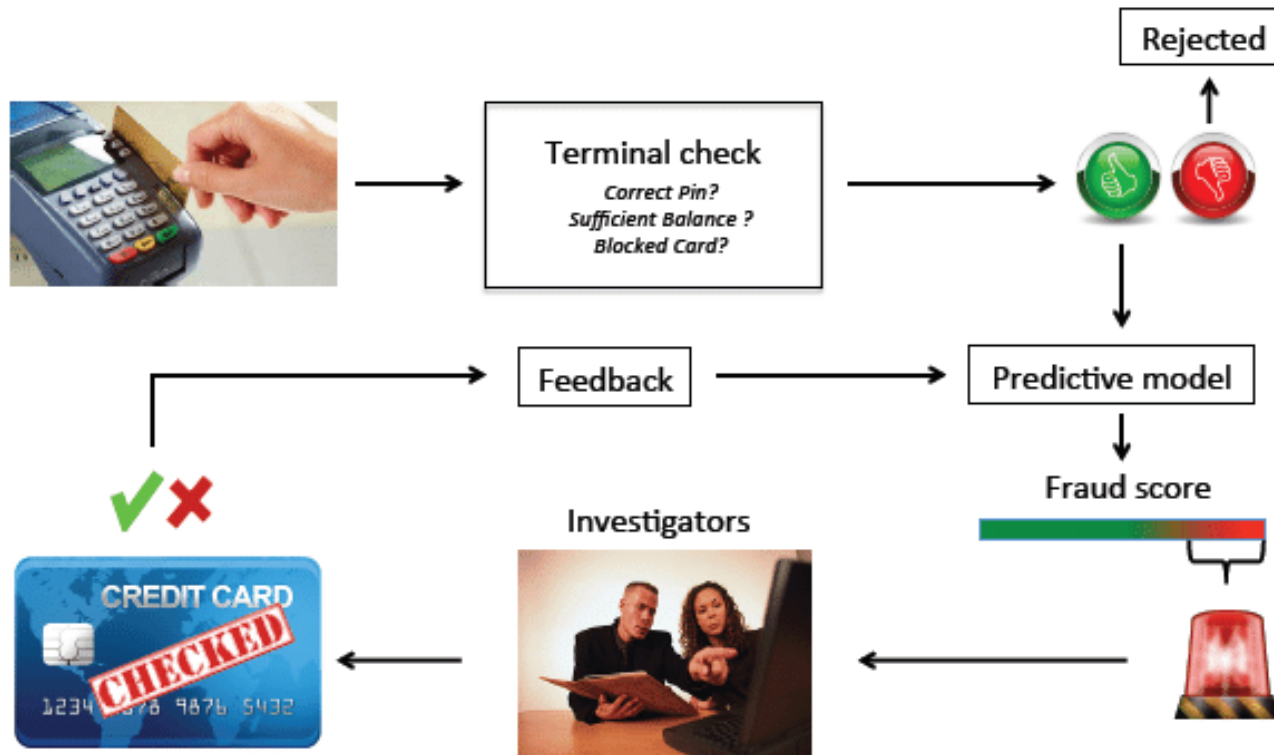


Visualizing a year of extreme weather



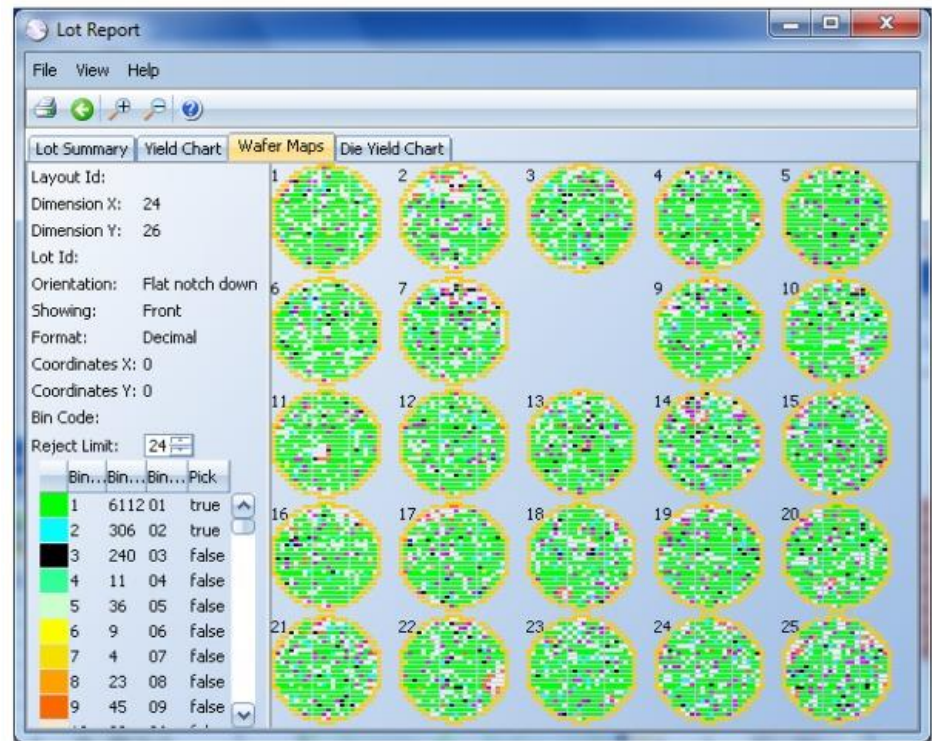
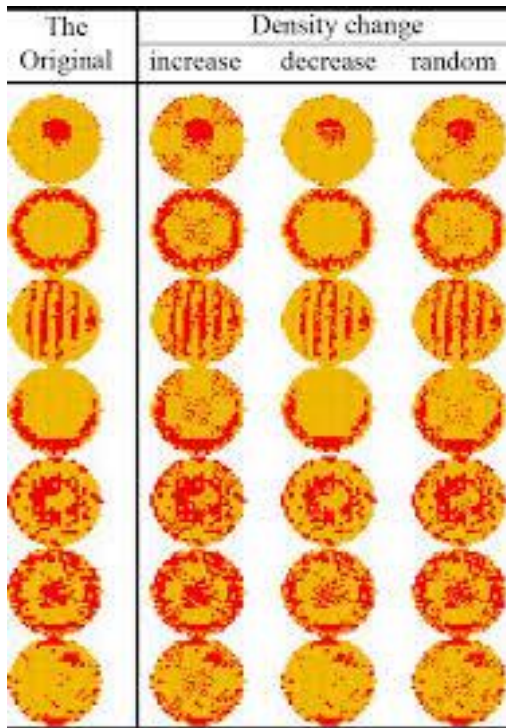
Anomaly (Novelty) Detection 사례

- 이상탐지 사례: 카드 이상거래 검출



Anomaly (Novelty) Detection 사례

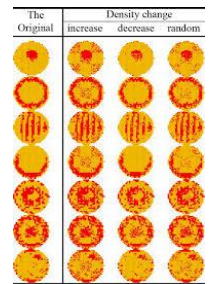
- 이상탐지 사례: 웨이퍼 품질 이상에 대한 이상탐지



Type of Abnormal Data

- **Global outlier** ★

- Object that deviates significantly from the rest of the data set
- Ex: Fault detection in manufacturing process
- Issues: Find an appropriate measurement of deviation

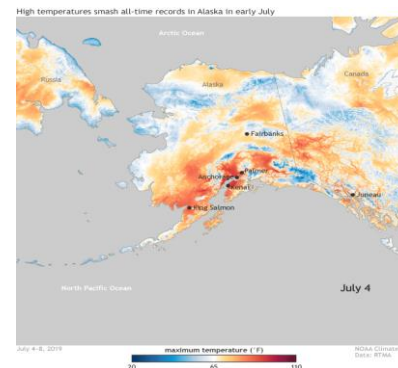


- **Contextual outlier**

- Object that deviate significantly based on a selected context
- Issue: How to define or formulate meaningful context



하와이에서의 40도

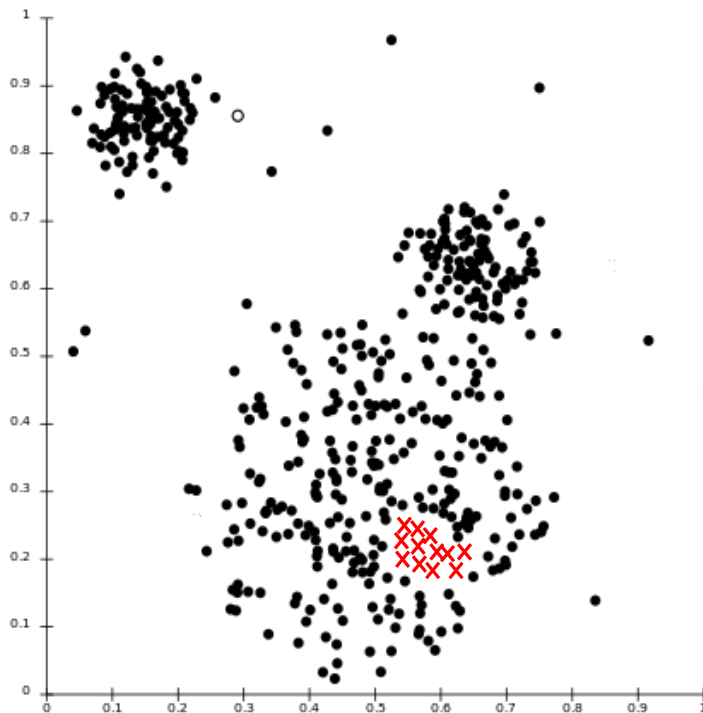


알래스카에서의 40도

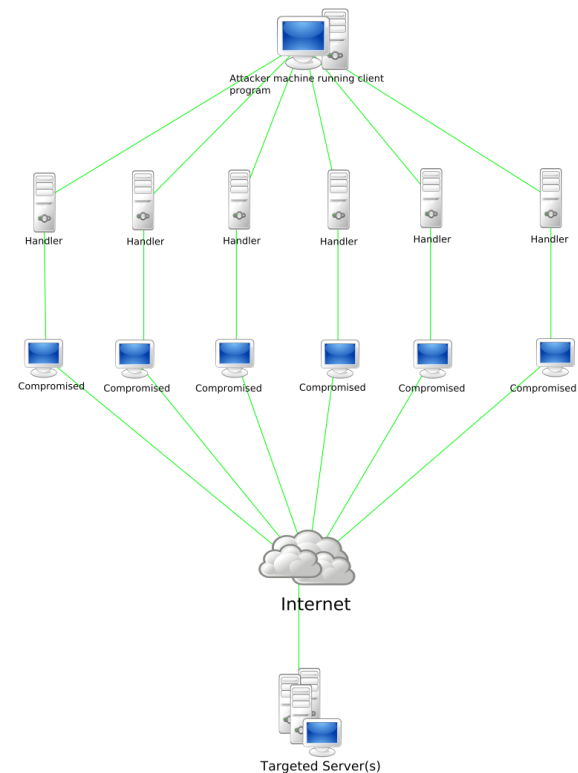
Type of Abnormal Data

- **Collective outlier**

- A subset of objects collectively deviate significantly from the whole data set, even if the individual data objects may not be outliers



A distributed Denial-of-service(DDoS) attack



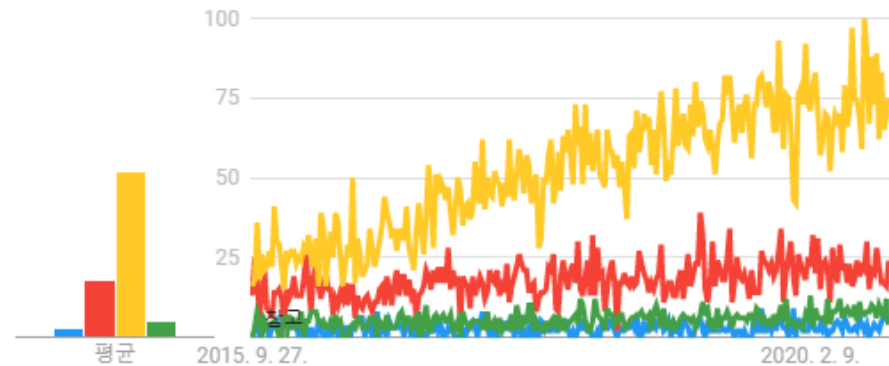
Anomaly (Novelty) Detection 뜻하는 용어

- Novelty Detection
- Outlier Detection
- Anomaly Detection
- One Class Classification

시간 흐름에 따른 관심도 변화

Google Trends

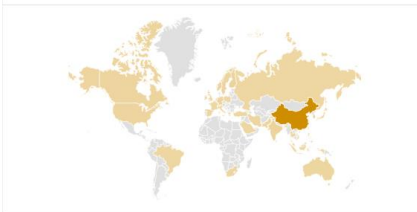
● Novelty Detection ● Outlier Detection ● Anomaly Detection
● One Class Classification



Anomaly Detection

지역별 관심도 ⓘ

지역 ▼ ⓘ ⬇ ⬅ ➡



Outlier Detection

지역별 관심도 ⓘ

지역 ▼ ⓘ ⬇ ⬅ ➡



Novelty Detection

지역별 관심도 ⓘ

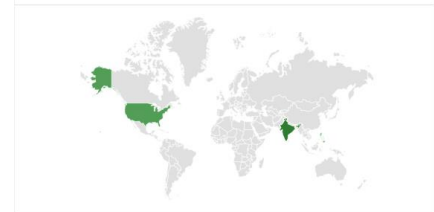
지역 ▼ ⓘ ⬇ ⬅ ➡



One Class Classification

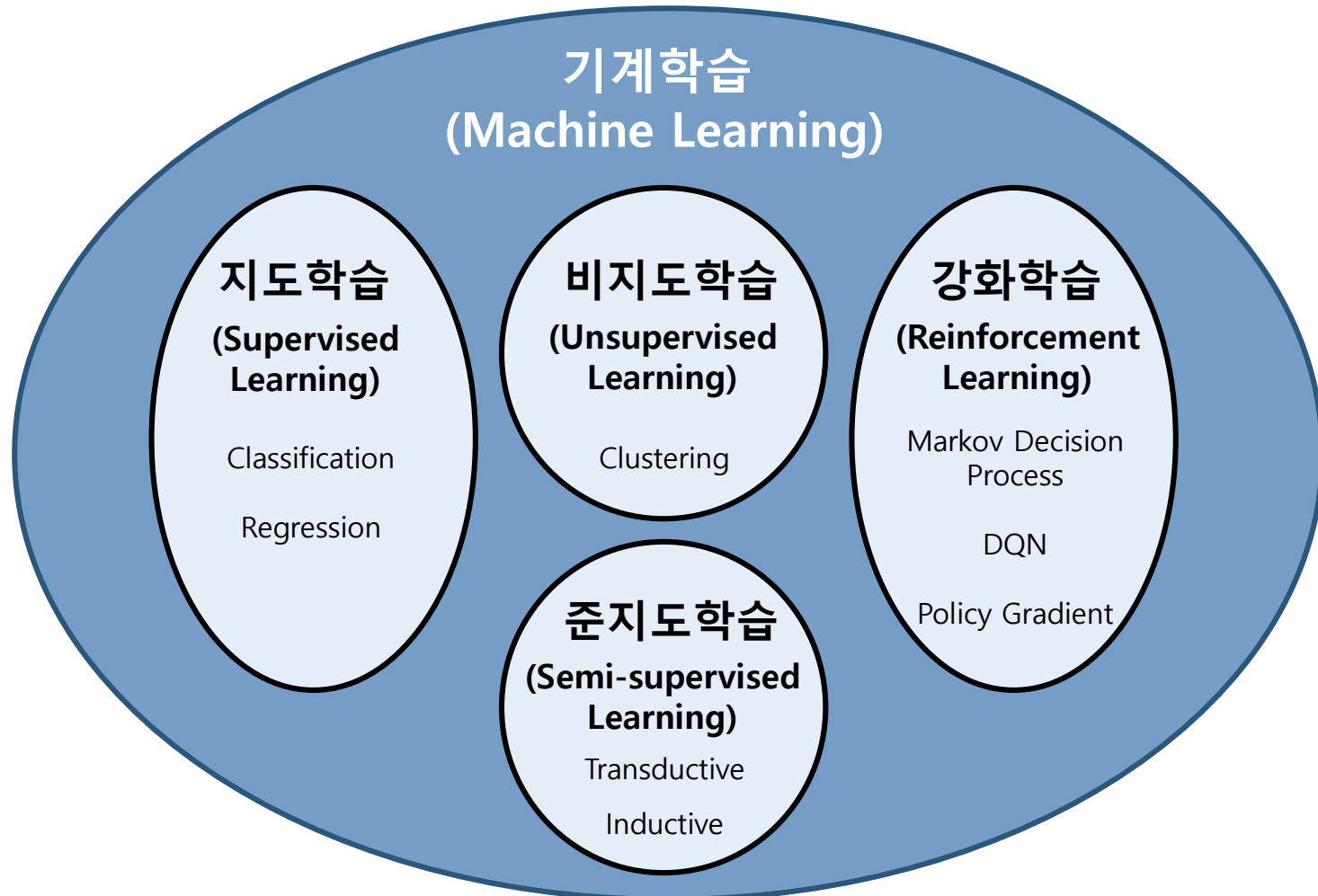
지역별 관심도 ⓘ

지역 ▼ ⓘ ⬇ ⬅ ➡



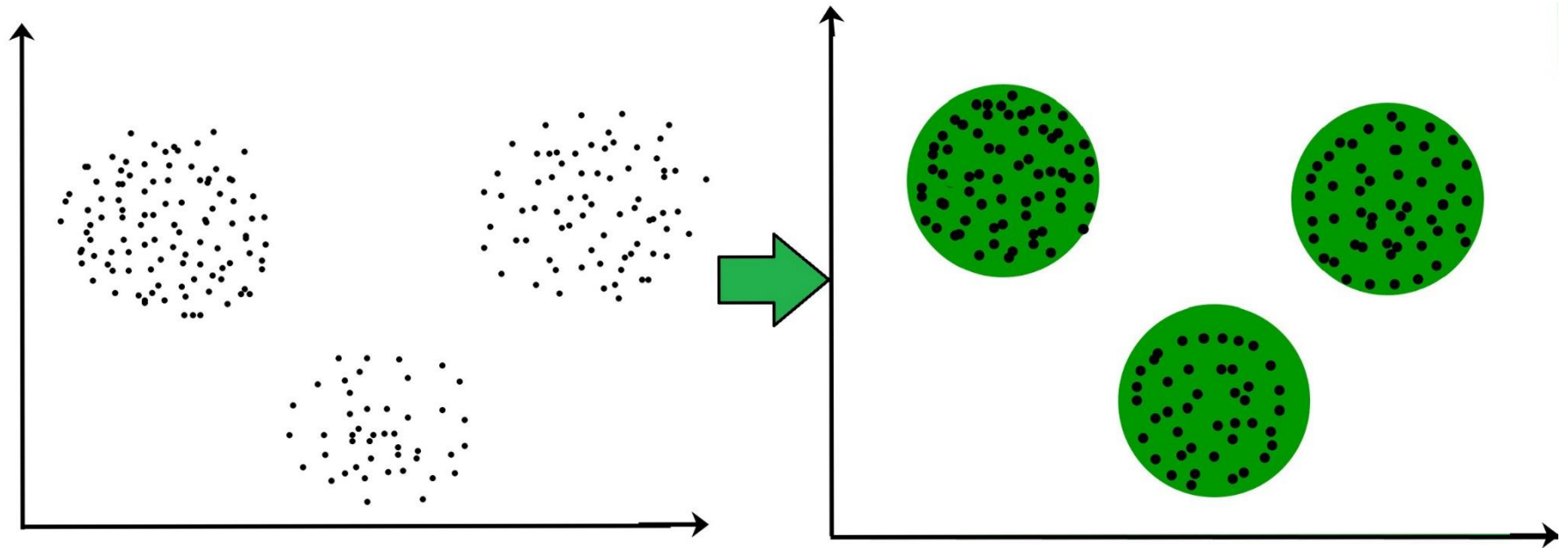
학습모델 구축 시, 문제 상황(Task) 별 분류 유형

■ 모델 유형



비지도 학습 (Unsupervised Learning)

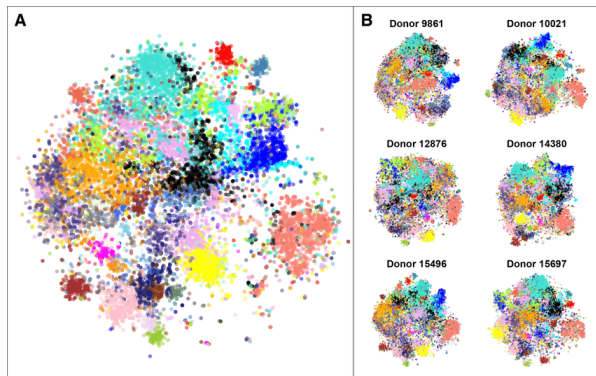
- 샘플에 대해서 군집(cluster)을 나누는 것이다.



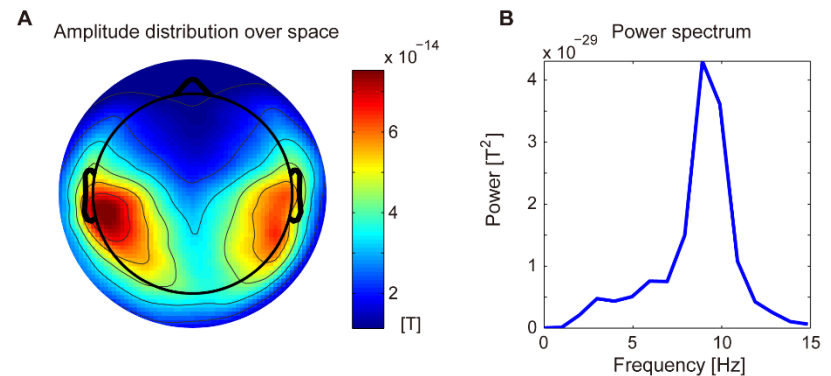
비지도 학습 (Unsupervised Learning)

- 비지도 학습에는 군집화만 있는 것이 아니다.

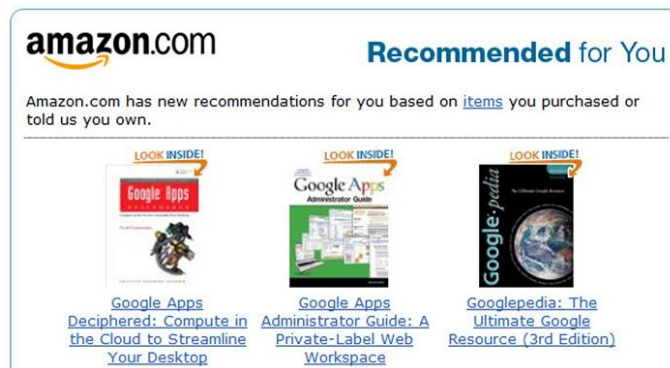
데이터 내재적 특징 탐색



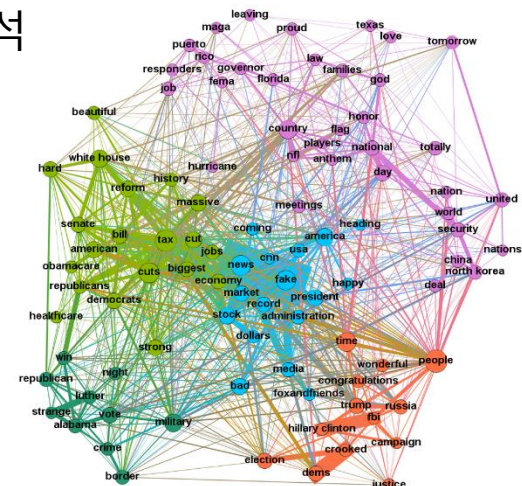
데이터 분포 추정



연관분석, 추천 시스템



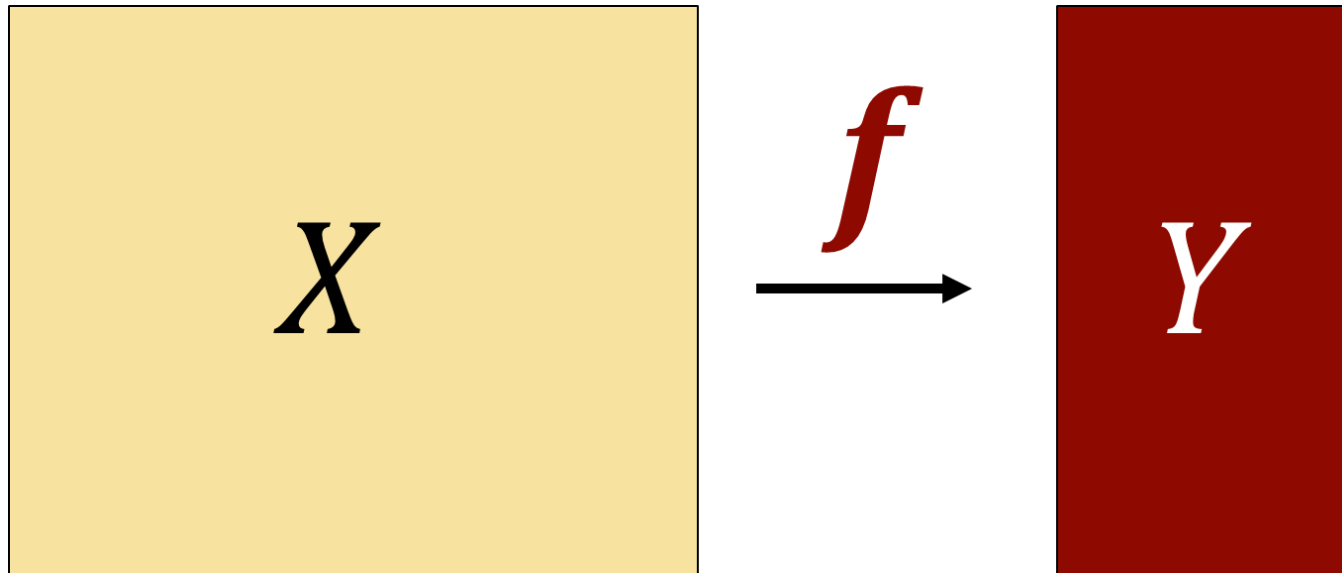
네트워크 분석



지도 학습 (Supervised Learning)

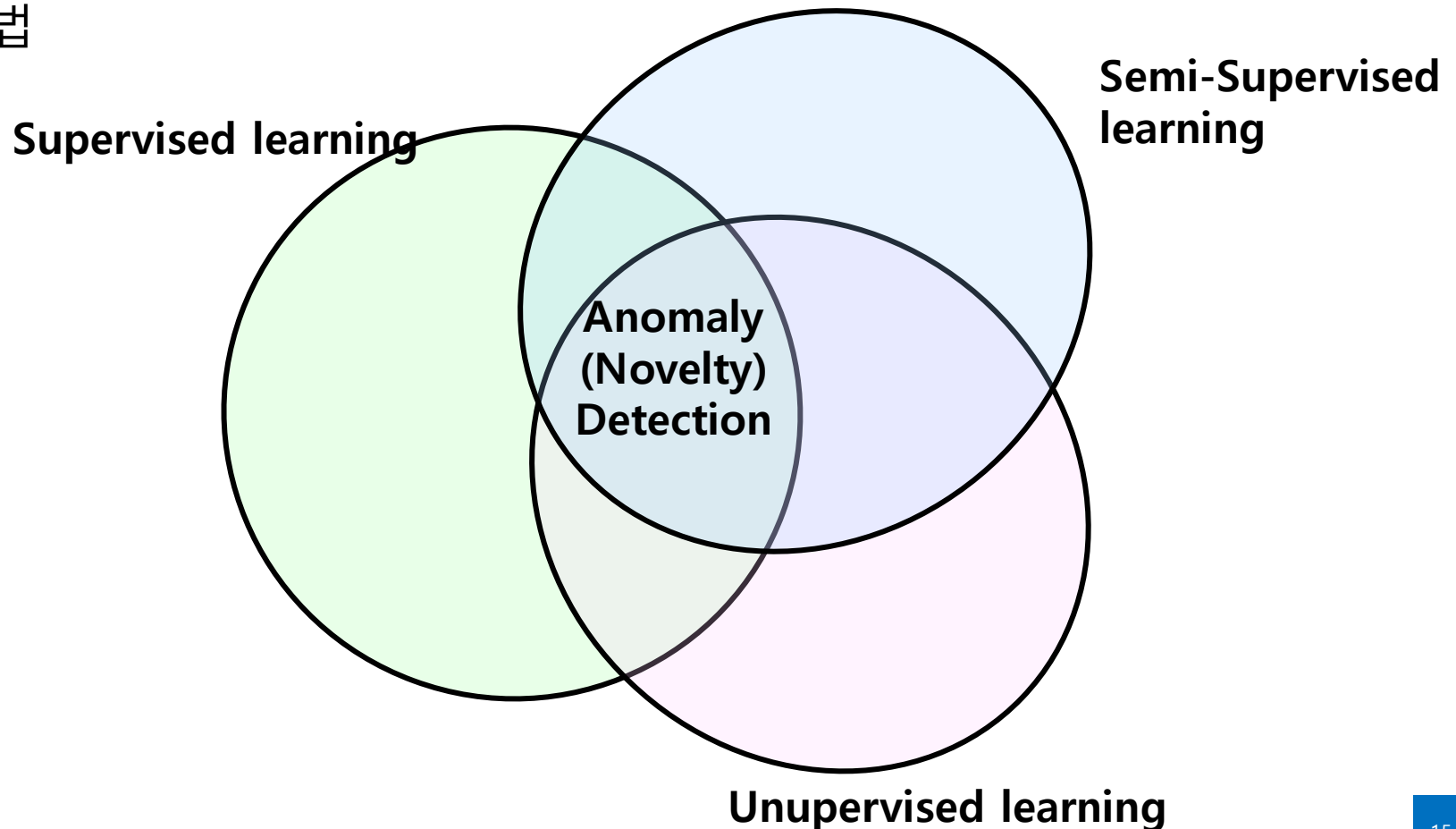
- 학습데이터로부터 $X \rightarrow Y$ 의 관계를 설명하는 함수 (f)를 찾는 방법론
- 종속변수 y 가 범주형이면 분류(classification), 연속형이면 회귀(regression)

$$f(x) \approx y$$



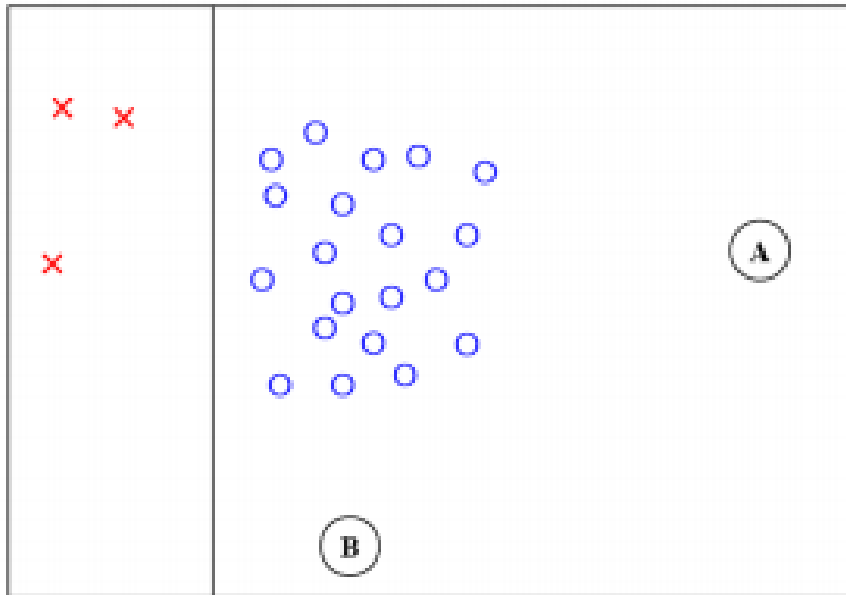
Anomaly (Novelty) Detection?

- 지도학습(Supervised learning) 또는 준지도학습(Semi-Supervised learning) 또는 비지도학습(Unsupervised learning) 에 속하는 분류 기법

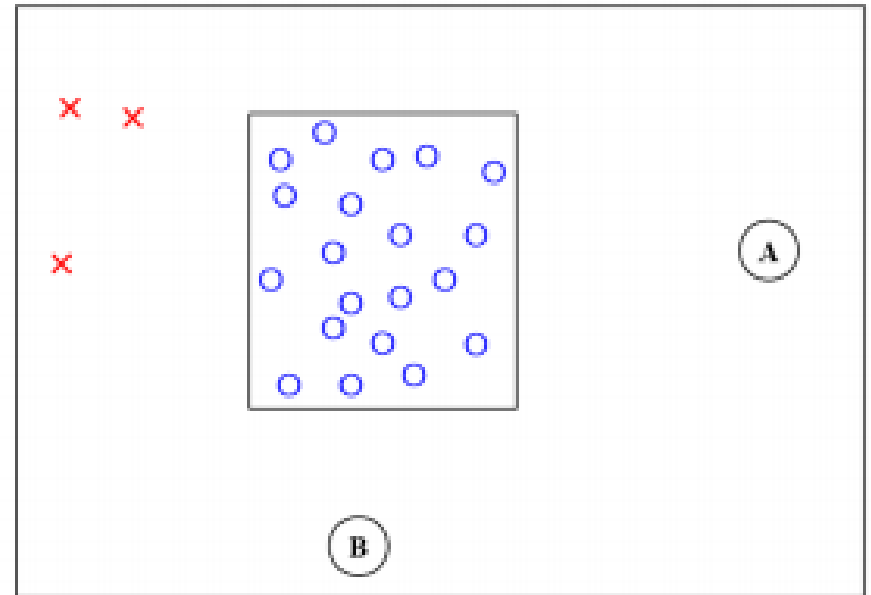


Anomaly (Novelty) Detection?

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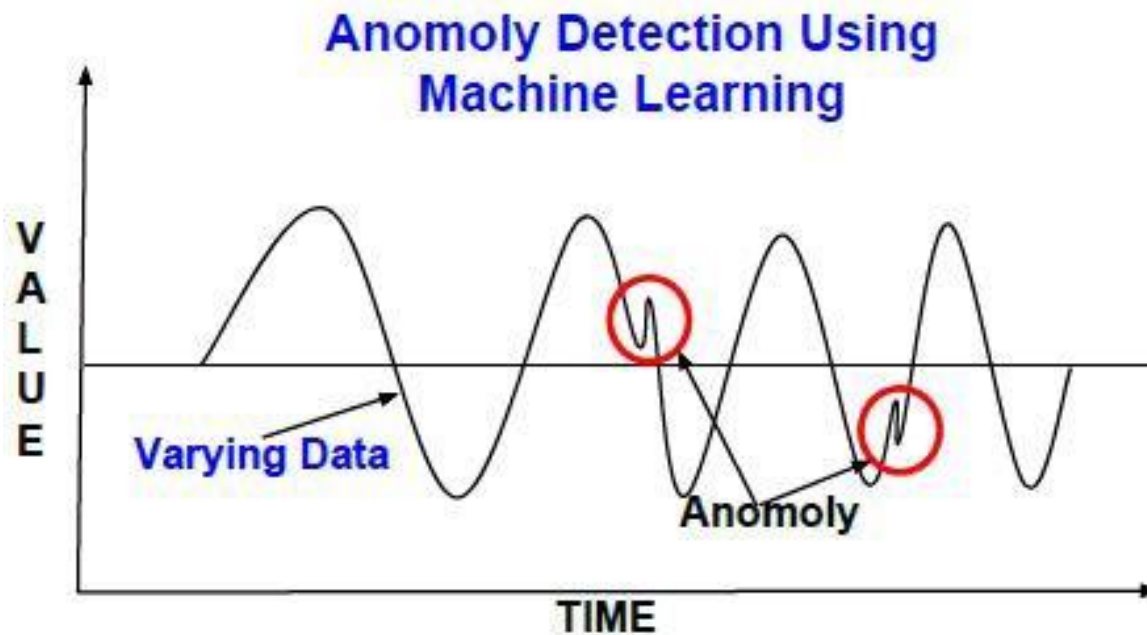
Binary classification



Anomaly detection

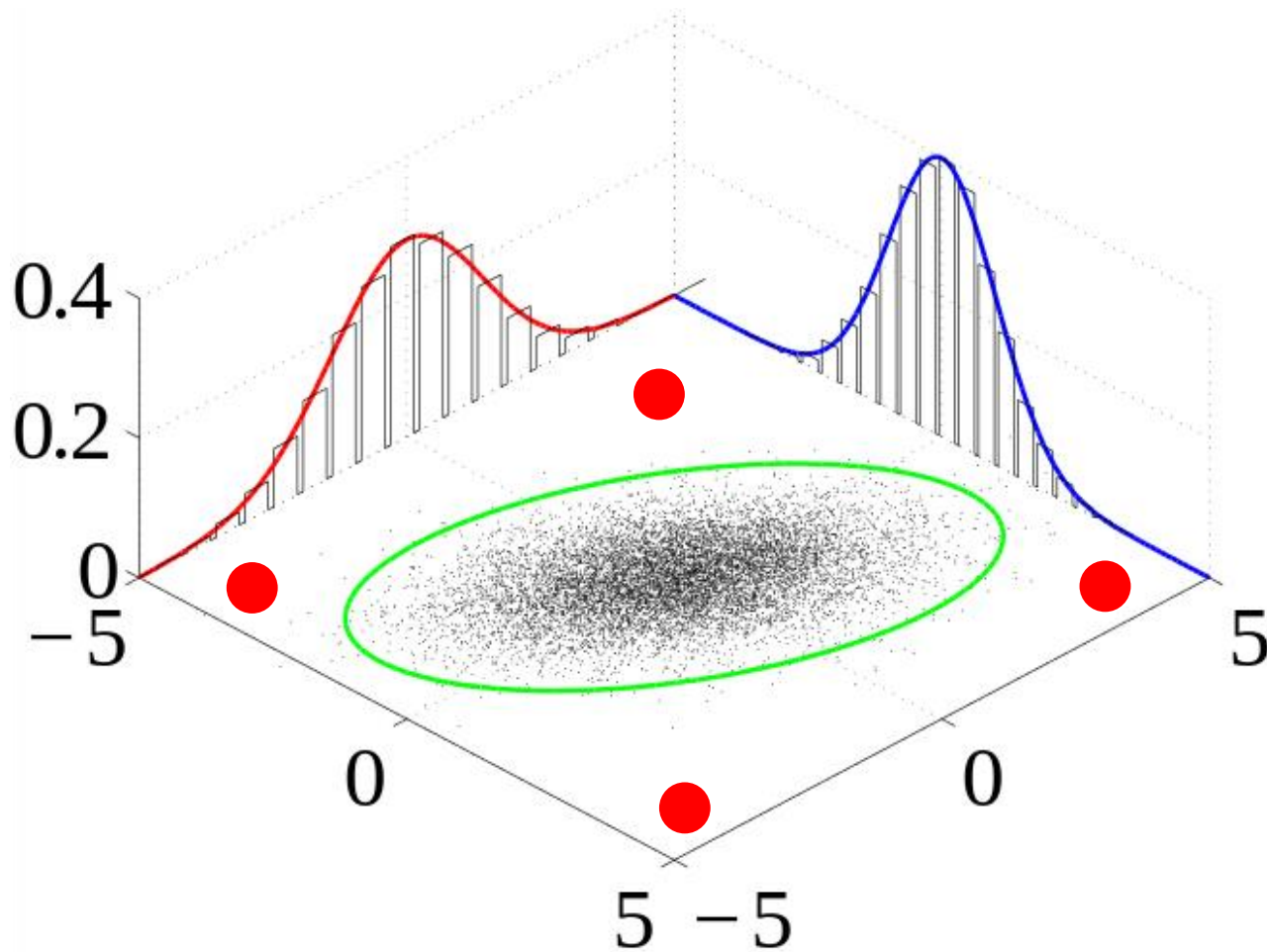
Anomaly (Novelty) Detection?

- 지도학습(Supervised learning) 또는 준지도학습(Semi-Supervised learning) 또는 비지도학습(Unsupervised learning) 에 속하는 분류 기법



Anomaly (Novelty) Detection?

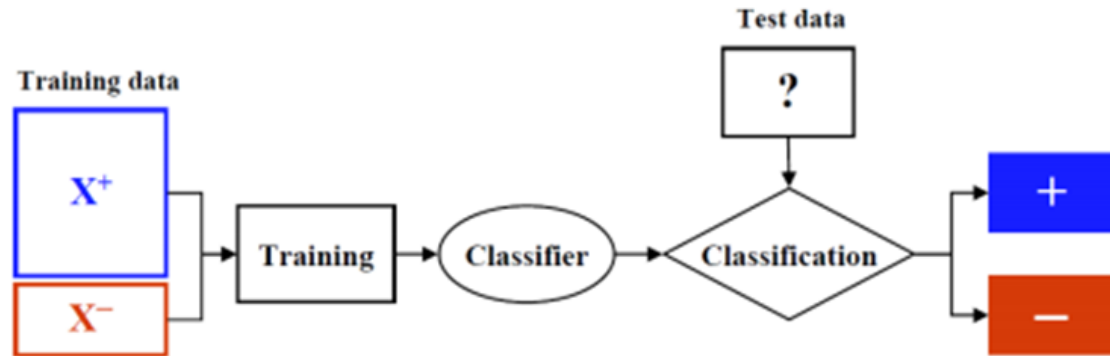
- 지도학습(Supervised learning) 또는 준지도학습(Semi-Supervised learning) 또는 비지도학습(Unsupervised learning) 에 속하는 분류 기법



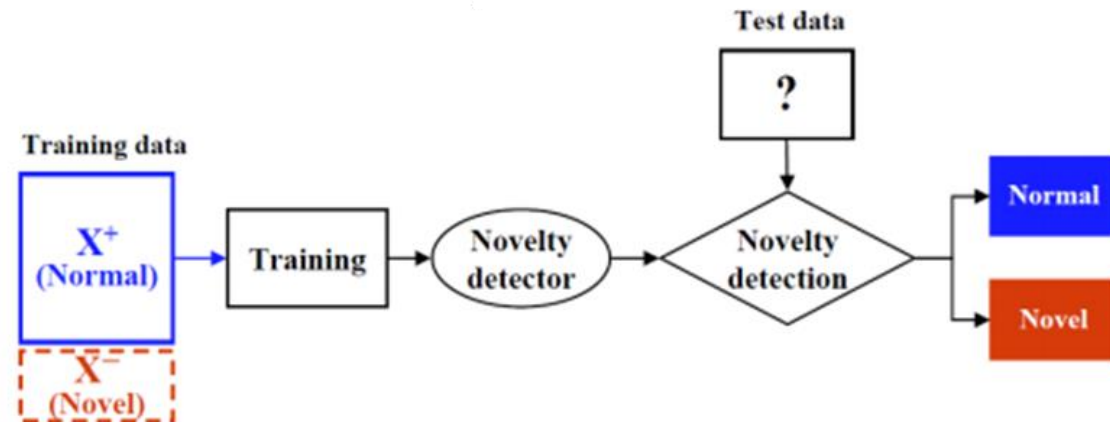
Anomaly (Novelty) Detection 접근방식

- '정상'으로 고려되는 데이터를 많이 갖고 있다.
- '정상'이 아니면 '이상'이다.

Classification

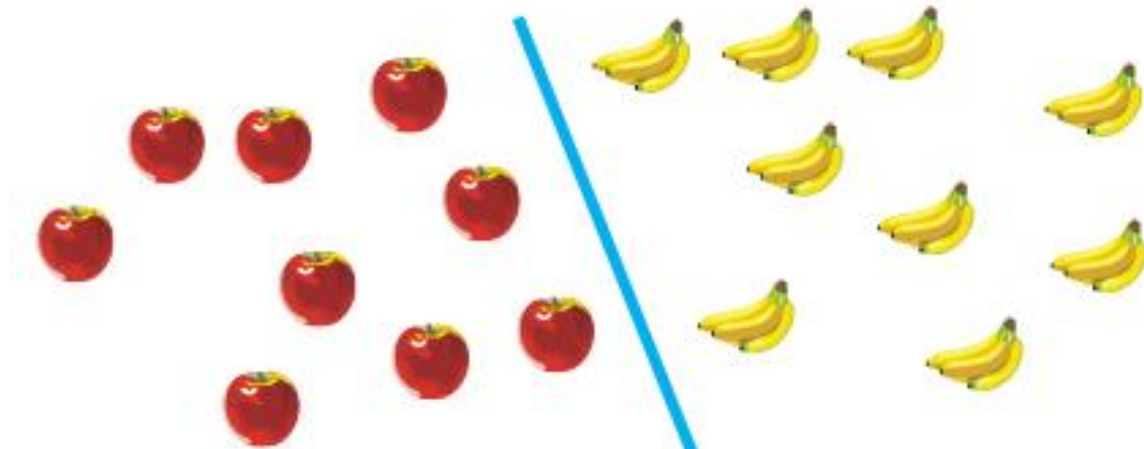


Anomaly Detection

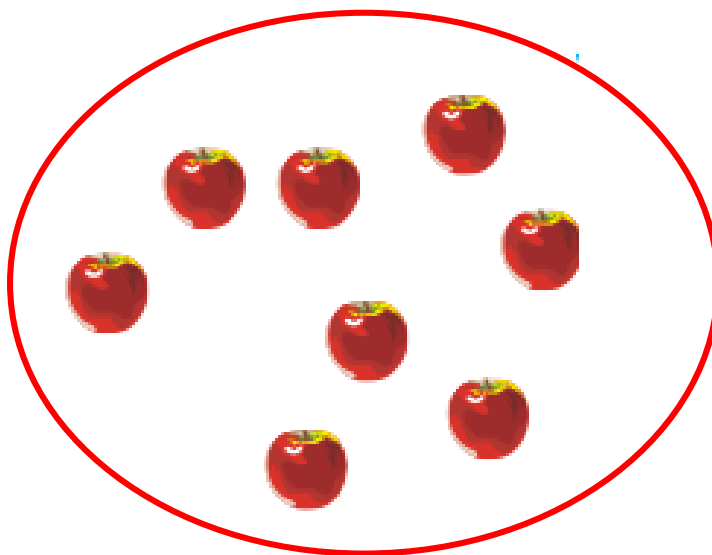


Anomaly (Novelty) Detection 접근방식

✓ Classification

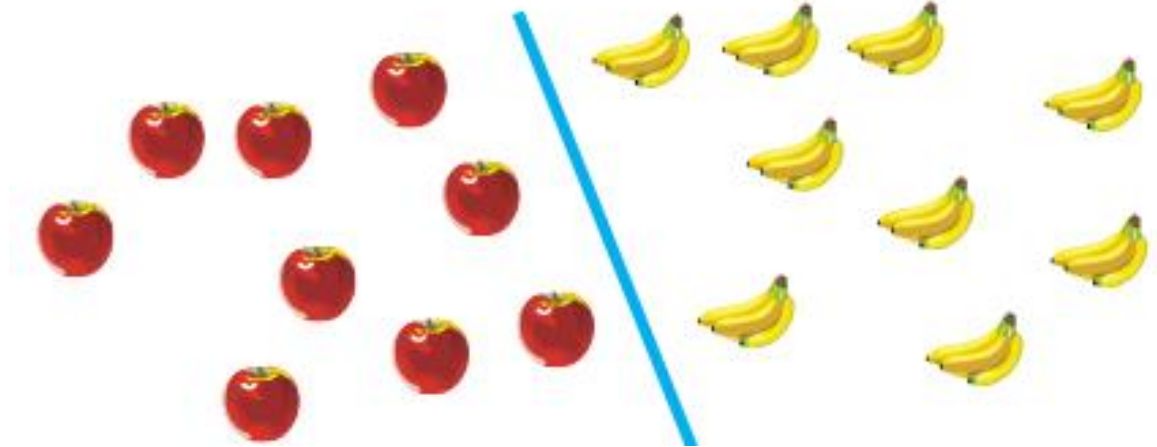


✓ Anomaly Detection

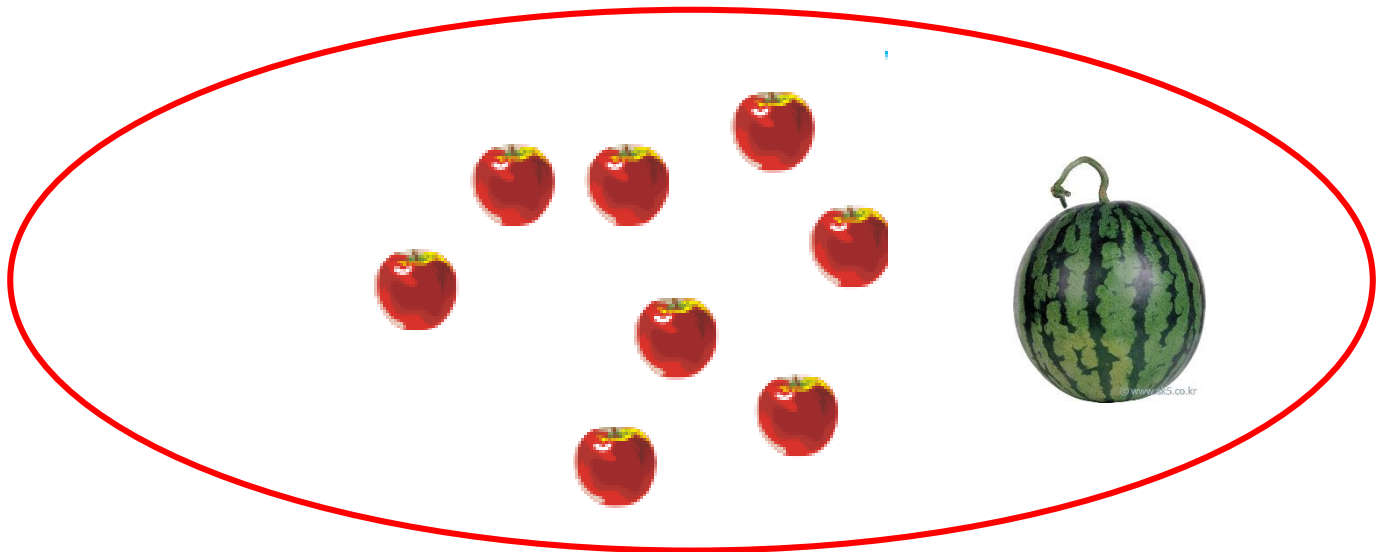


Anomaly (Novelty) Detection 접근방식

✓ Classification

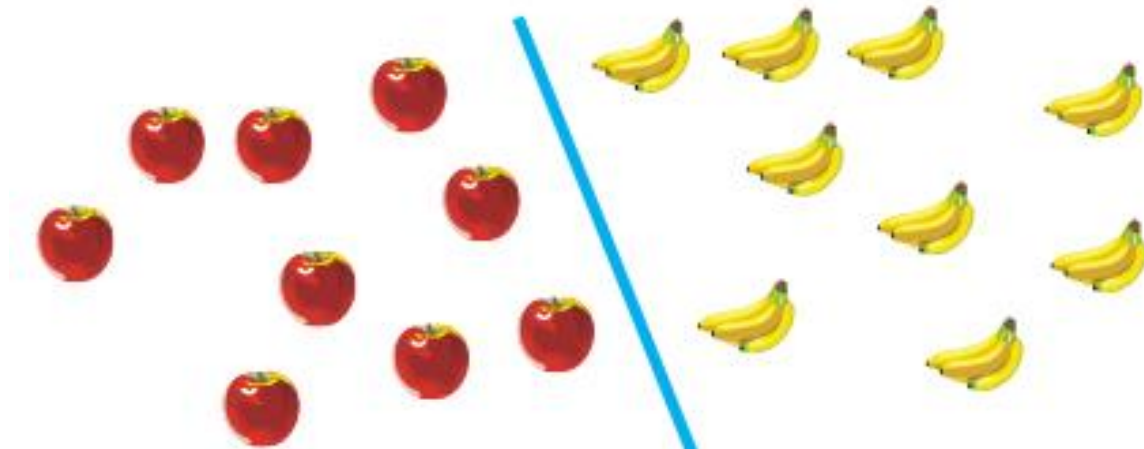


✓ Anomaly Detection

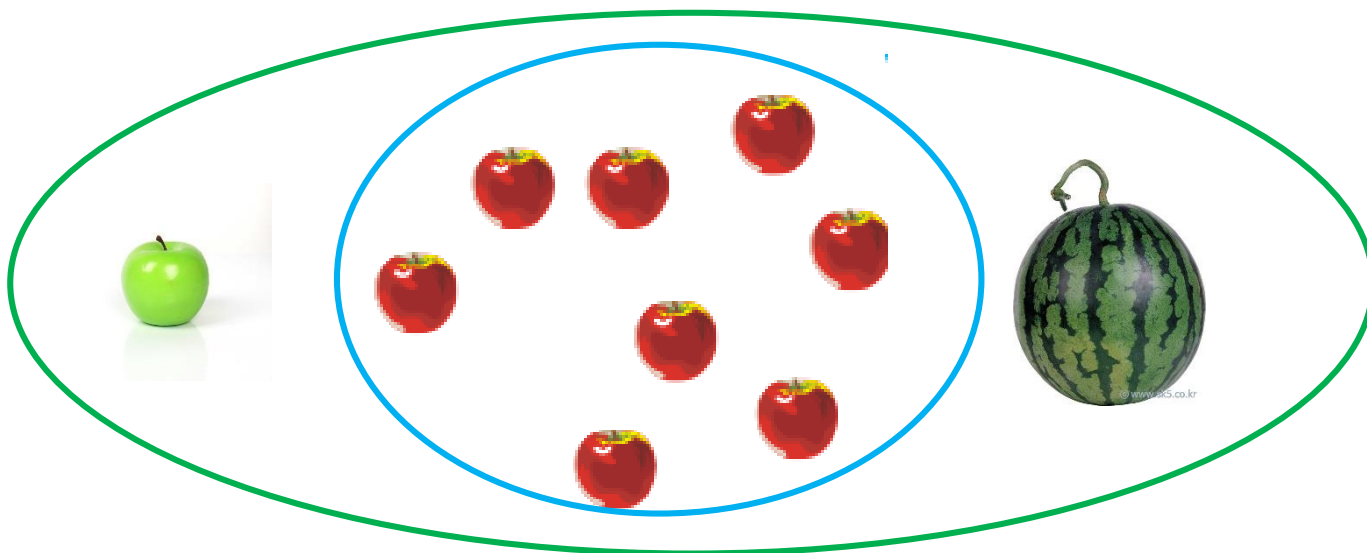


Anomaly (Novelty) Detection 접근방식

✓ Classification

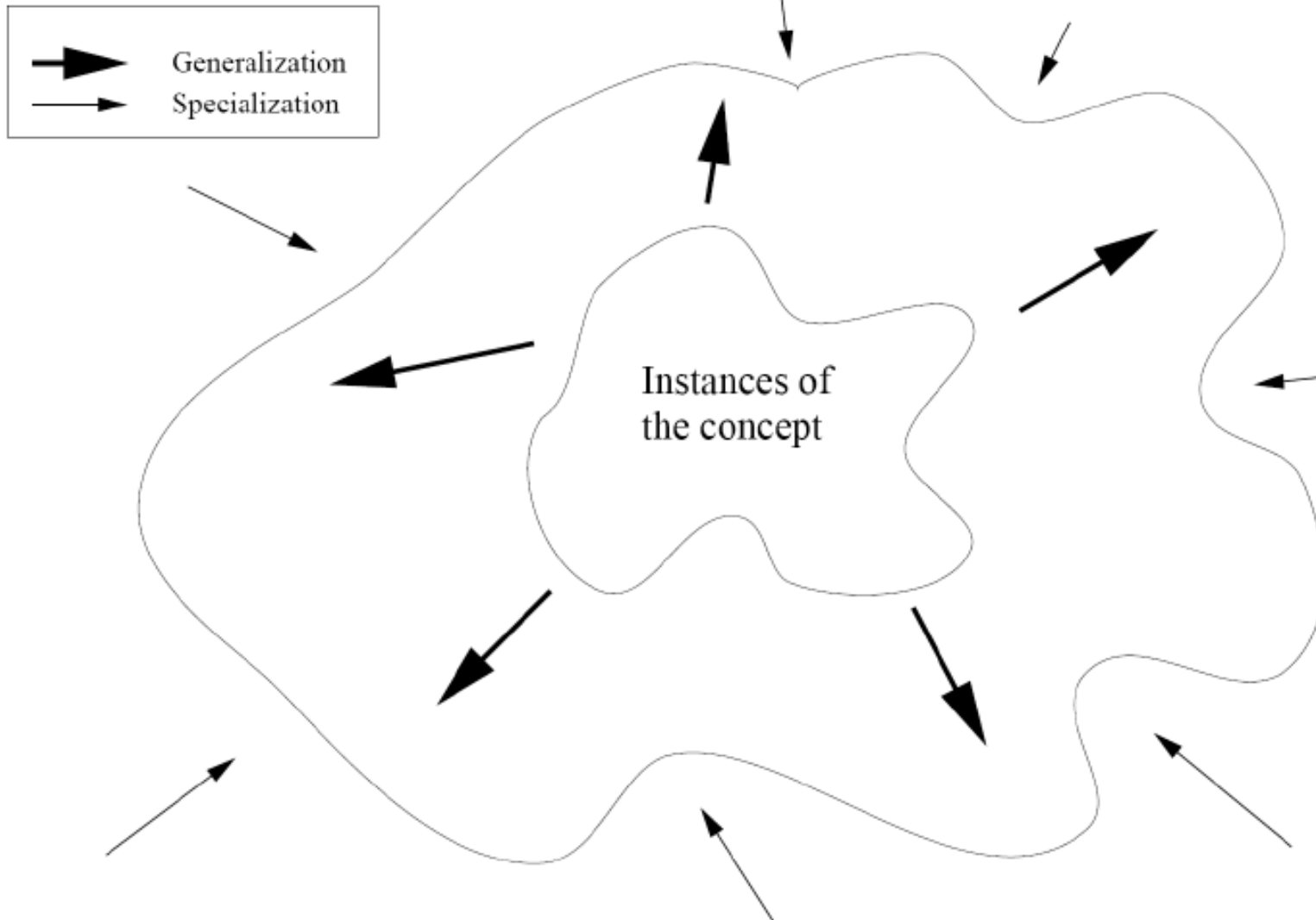


✓ Anomaly Detection



Generalization vs. Specialization

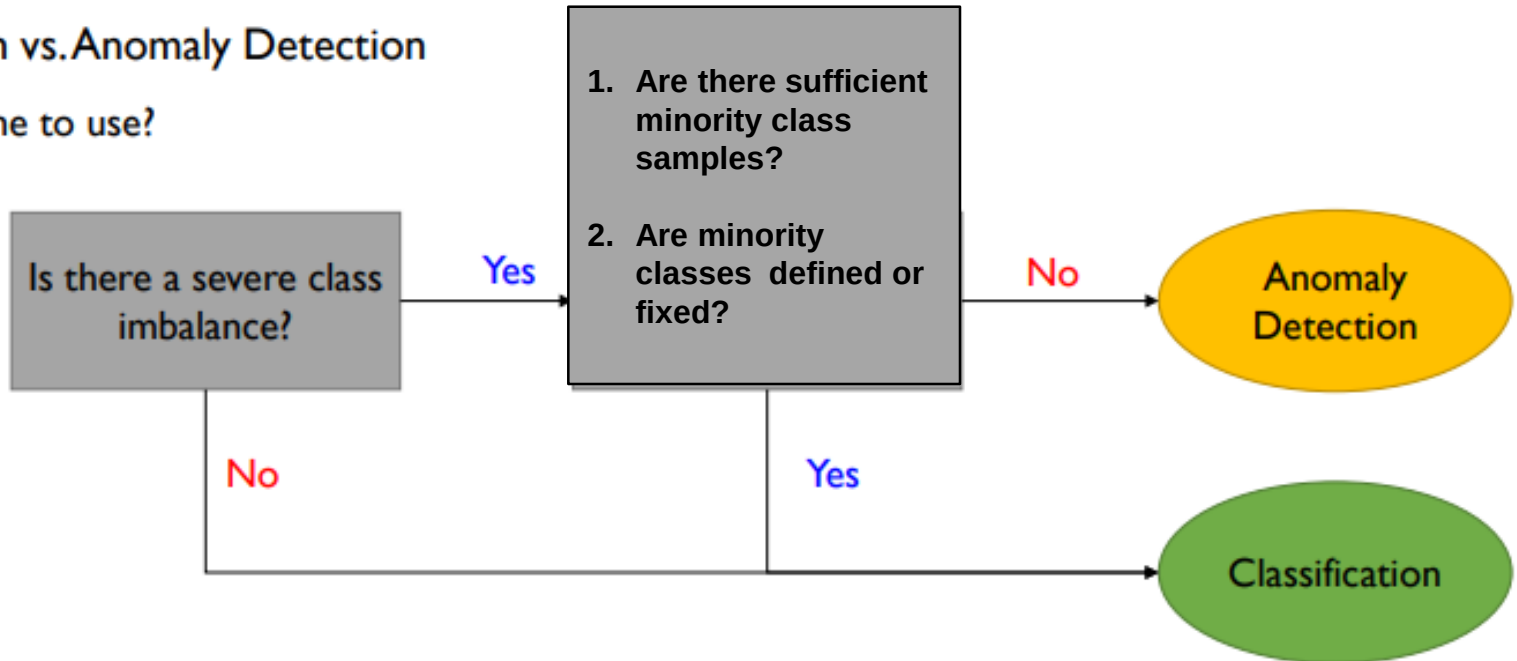
- '정상'에 대한 boundary를 정하자.



Anomaly (Novelty) Detection 접근방식

- Classification vs. Anomaly Detection

✓ Which one to use?



Challenges in Anomaly (Novelty) Detection

- Modeling normal objects and outliers properly

- ✓ The border between normal and outlier objects is often a gray area

- Application-specific outlier detection

- ✓ Choice of distance measure among objects and the model of relationship among objects are often application-dependent
- ✓ E.g., clinic data: a small deviation could be an outlier; while in marketing analysis, larger fluctuations

- Understandability

- ✓ Understand why these are outliers: Justification of the detection
- ✓ Specify the degree of an outlier: the unlikelihood of the object being generated by a normal mechanism

Challenges in Anomaly (Novelty) Detection

- Performance Measures

- ✓ Confusion matrix for novelty detection

		Predicted class	
		Abnormal	Normal
Actual class	Abnormal	A	B
	Normal	C	D

- ✓ Performance measures when the cut-off (threshold) is set

Metric	Description
Detection Rate	(Identified as abnormal)/(Actually abnormal) = $A/(A+B)$
False Rejection Rate (FRR)	(Rejected as abnormal)/(Actually normal) = $C/(C+D)$
False Acceptance Rate (FAR)	(Accepted as normal)/(Actually abnormal) = $B/(A+B)$

Challenges in Anomaly (Novelty) Detection

- Performance Measures

- ✓ Confusion matrix for novelty detection

		Predicted class	
		Abnormal	Normal
Actual class	Abnormal	A	B
	Normal	C	D

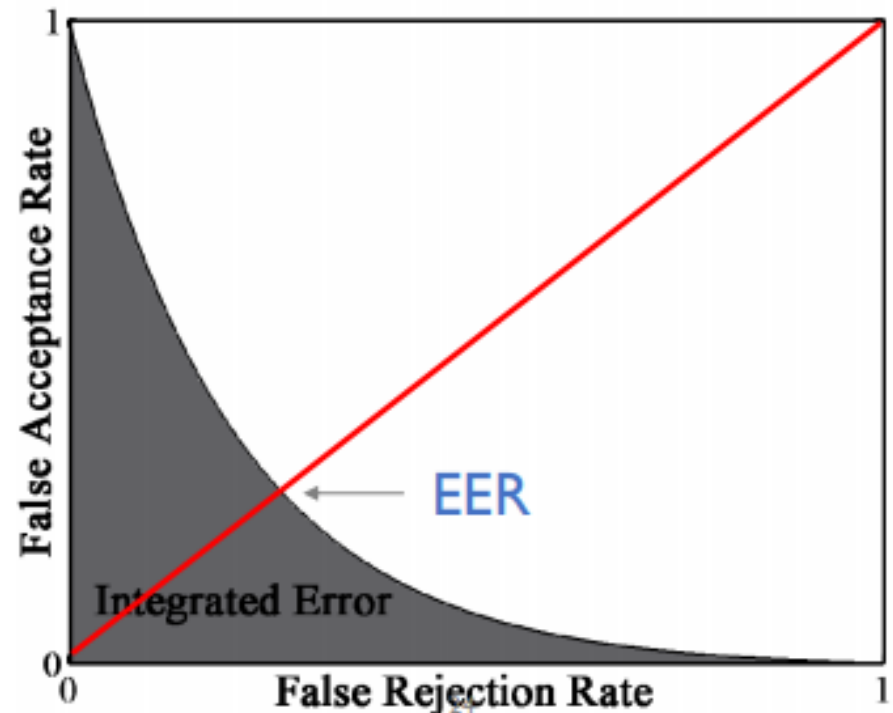
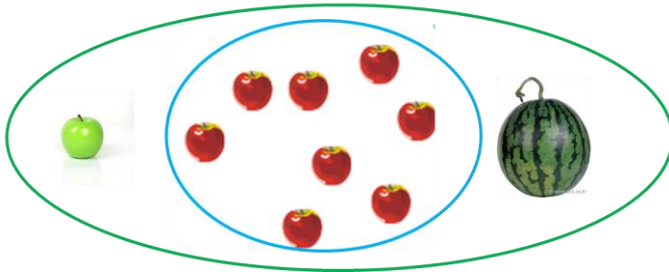
- ✓ Performance measures when the cut-off (threshold) is set

Metric	Description
Detection Rate	(Identified as abnormal)/(Actually abnormal) = $A/(A+B)$
False Rejection Rate (FRR)	(Rejected as abnormal)/(Actually normal) = $C/(C+D)$
False Acceptance Rate (FAR)	(Accepted as normal)/(Actually abnormal) = $B/(A+B)$

Performance Measures

- To evaluate an intrinsic performance of novelty detection algorithms
 - ✓ Equal error rate (EER): Error rate where the FAR and FRR are the same
 - ✓ Integrated Error (IE): the area under the FRR-FAR curve
 - AUROC for classification: the higher the better
 - IE for anomaly detection: the lower the better

✓ Anomaly Detection

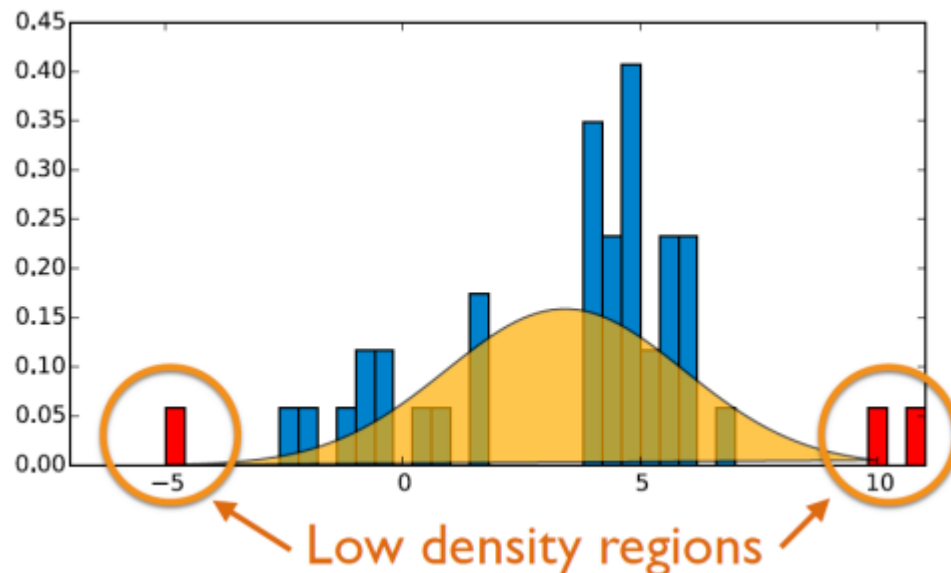


Density Estimation-based Approach

“Observations that deviate so much from other observations as to arouse suspicions that they were generated by a different mechanism (Hawkins, 1980)”

“Instances that their true probability density is very low (Harmeling et al., 2006)”

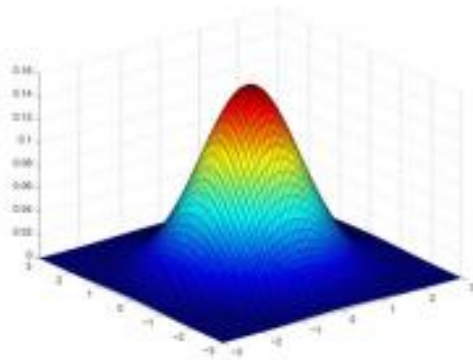
- Purpose
 - ✓ Estimate the data-driven density function
 - ✓ If a new instance has a low probability according the trained density function, it will be identified as novel



Density Estimation-based Approach

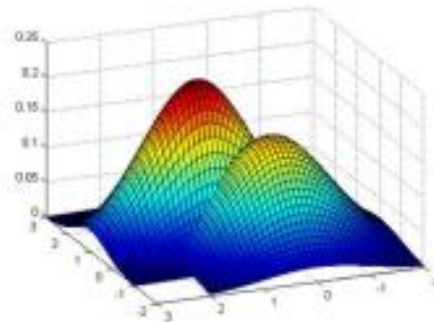
- Purpose
 - ✓ Estimate the data-driven density function
 - ✓ If a new instance has a low probability according the trained density function, it will be identified as novel

Gaussian Density Estimation



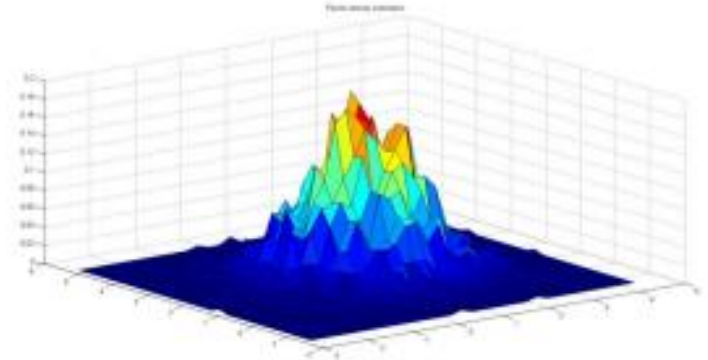
Number of modals
= 1

Mixture of Gaussian Density Estimation



1 <
Number of modals
< Number of instances

Kernel Density Estimation



Number of modals
= Number of instances

Density Estimation-based Approach

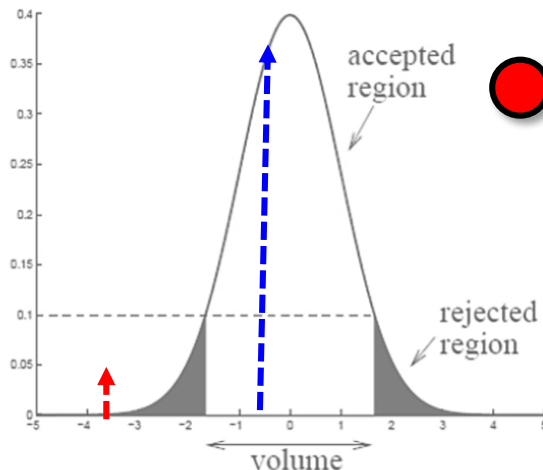
- **Gaussian (Parametric) Density Estimation**

- ✓ Assume that the observed data are drawn from a Gaussian distribution

$x \sim \text{Univariate (Gaussian) Normal Distribution } (\mu, \sigma^2)$

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x-\mu)^2}{2\sigma^2} \right\}$$

● 관측치 #1



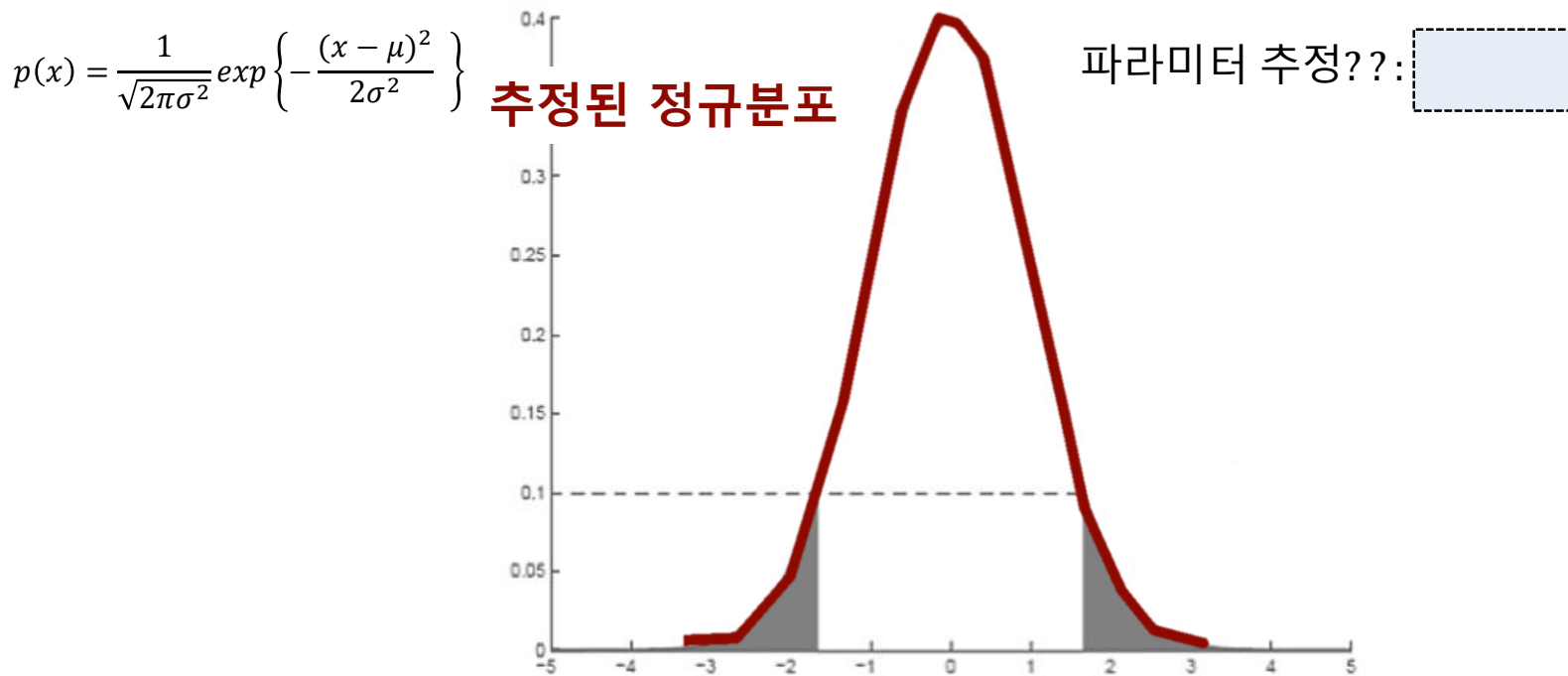
● 관측치 #2

Density Estimation-based Approach

- Gaussian Density Estimation

- ✓ Step 1. 학습모델 구축 I (정규분포 가정)

→ 정상데이터만을 학습데이터로 활용, 정상데이터에 대한 정규분포 추정



Density Estimation-based Approach

- Gaussian Density Estimation

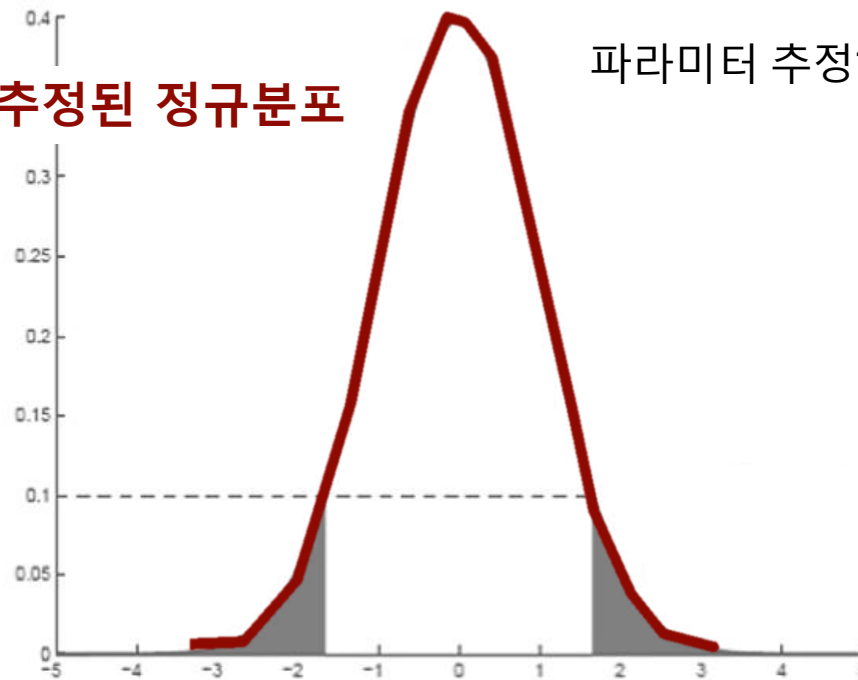
- ✓ Step 1. 학습모델 구축 I (정규분포 가정)

→ 정상데이터만을 학습데이터로 활용, 정상데이터에 대한 정규분포 추정

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

추정된 정규분포

파라미터 추정??:



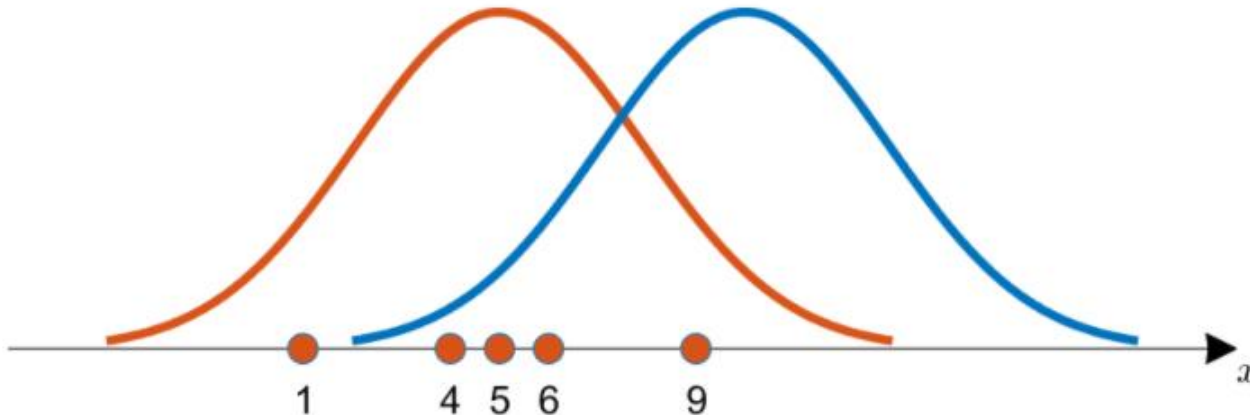
Density Estimation-based Approach

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} \quad \text{파라미터 추정??: } \mu, \sigma^2$$

How ?? → Maximum Likelihood Estimation (MLE)

$$p(x_i|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x_i-\mu)^2}{2\sigma^2}\right\} \quad \text{Likelihood} = \prod_{i=1}^n p(x_i|\mu, \sigma^2)$$

$$\text{Likelihood} = \prod_{i=1}^n p(x_i|\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x_i-\mu)^2}{2\sigma^2}\right\}$$



Density Estimation-based Approach

$$Likelihood = \prod_{i=1}^n p(x_i | \mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x_i - \mu)^2}{2\sigma^2} \right\}$$

$$\text{maximize}_{\hat{\mu}, \hat{\sigma}^2} Likelihood = \text{maximize}_{\hat{\mu}, \hat{\sigma}^2} \prod_{i=1}^n \frac{1}{\sqrt{2\pi\hat{\sigma}^2}} \exp \left\{ -\frac{(x_i - \hat{\mu})^2}{2\hat{\sigma}^2} \right\}$$

$$\rightarrow \text{Log} \prod_{i=1}^n \frac{1}{\sqrt{2\pi\hat{\sigma}^2}} \exp \left\{ -\frac{(x_i - \hat{\mu})^2}{2\hat{\sigma}^2} \right\}, \text{Log Likelihood}$$

$$= \sum_{i=1}^n \left\{ \text{Log} \frac{1}{\sqrt{2\pi\hat{\sigma}^2}} + \text{Log} \exp \left\{ -\frac{(x_i - \hat{\mu})^2}{2\hat{\sigma}^2} \right\} \right\}$$

$$= \sum_{i=1}^n \left\{ \text{Log} \frac{1}{\sqrt{2\pi\hat{\sigma}^2}} - \frac{(x_i - \hat{\mu})^2}{2\hat{\sigma}^2} \right\}$$

Density Estimation-based Approach

$$\text{maximize}_{\hat{\mu}, \hat{\sigma}^2} \text{Likelihood} := \text{maximize}_{\hat{\mu}, \hat{\sigma}^2} \sum_{i=1}^n \left\{ \text{Log} \frac{1}{\sqrt{2\pi\hat{\sigma}^2}} - \frac{(x_i - \hat{\mu})^2}{2\hat{\sigma}^2} \right\}$$

$$\gamma = \frac{1}{\hat{\sigma}^2}$$

$$L = \sum_{i=1}^n \left\{ -\frac{1}{2} \text{Log } 2\pi + \frac{1}{2} \text{Log } \gamma - \gamma \frac{(x_i - \hat{\mu})^2}{2} \right\}$$

$$\frac{\partial L}{\partial \hat{\mu}} = \gamma \sum_{i=1}^n (x_i - \hat{\mu}) = 0 \rightarrow$$

$$\therefore \hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

$$\frac{\partial L}{\partial \hat{\gamma}} = \sum_{i=1}^n \left\{ \frac{1}{2} \frac{1}{\gamma} - \frac{(x_i - \hat{\mu})^2}{2} \right\} = 0$$

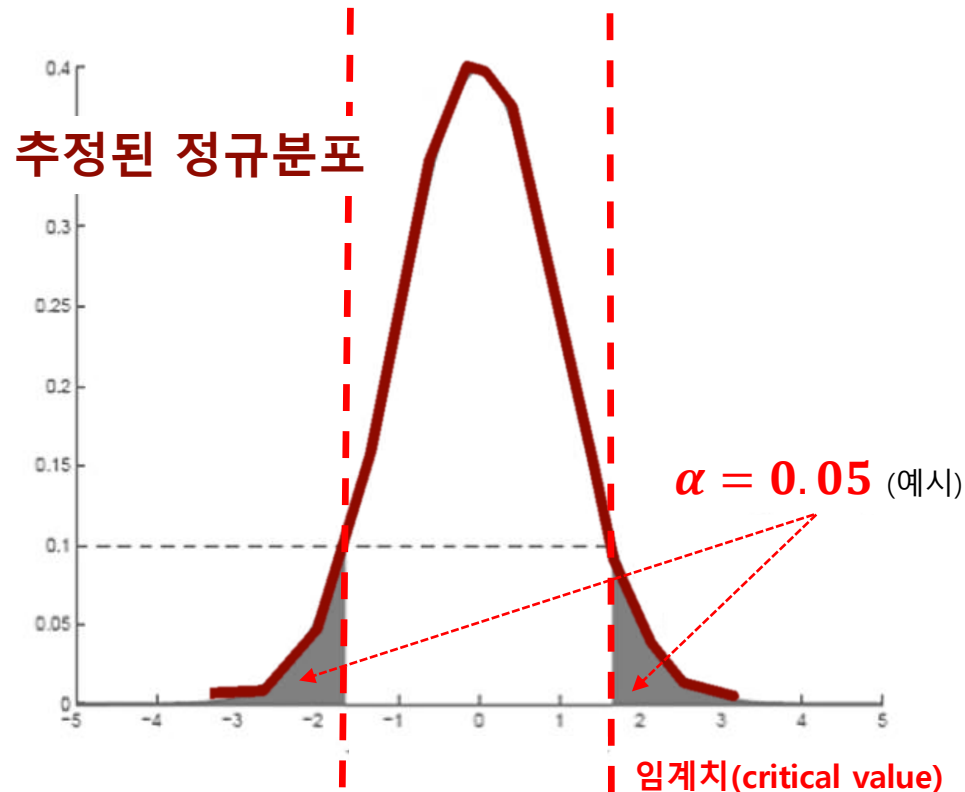
$$\therefore \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2$$

Density Estimation-based Approach

- Gaussian Density Estimation

- ✓ Step 2. Threshold 구축 II (임계치 설정)

→ Anomaly Detection task에 따라, 적정 수준의 임계치를 정하기

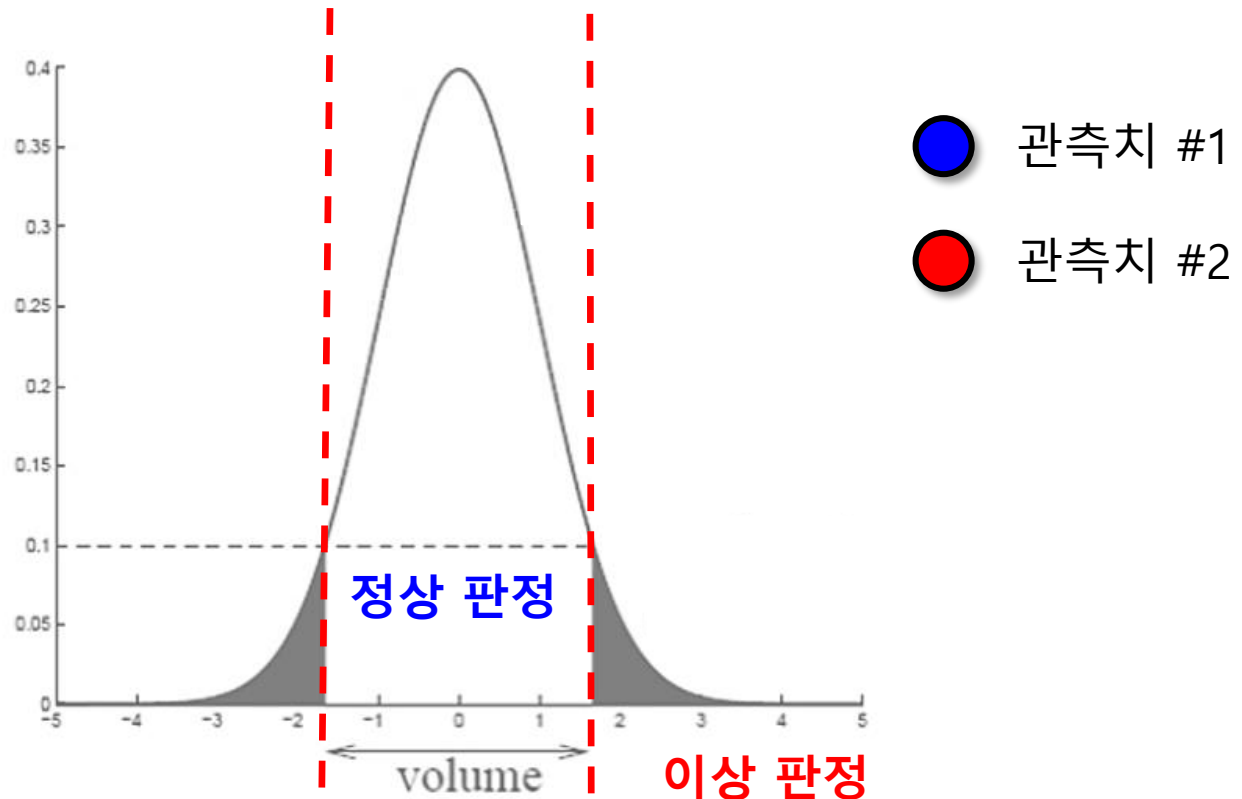


Density Estimation-based Approach

- Gaussian Density Estimation

- ✓ Step 3. 새 관측치에 대해 이상 유무를 판단

- 임계치(with 모델)를 기준으로 새 관측치의 정상/이상 판단



Density Estimation-based Approach

- **Gaussian (Parametric) Density Estimation**

- ✓ Assume that the observed data are drawn from a Gaussian distribution

$X = [x_1, x_2, \dots, x_k] \sim \text{Multivariate (Gaussian) Normal Distribution } (\mu, \Sigma)$

Univariate

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

Multivariate

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left[\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)\right]$$

$$\Sigma = \frac{1}{n} \sum_{\mathbf{x}_i \in \mathbf{X}^+} (\mathbf{x}_i - \mu)(\mathbf{x}_i - \mu)^T \text{ (covariance matrix)}$$

$$\mu = \frac{1}{n} \sum_{\mathbf{x}_i \in \mathbf{X}^+} \mathbf{x}_i \text{ (mean vector)}$$

Density Estimation-based Approach

- **Gaussian (Parametric) Density Estimation**

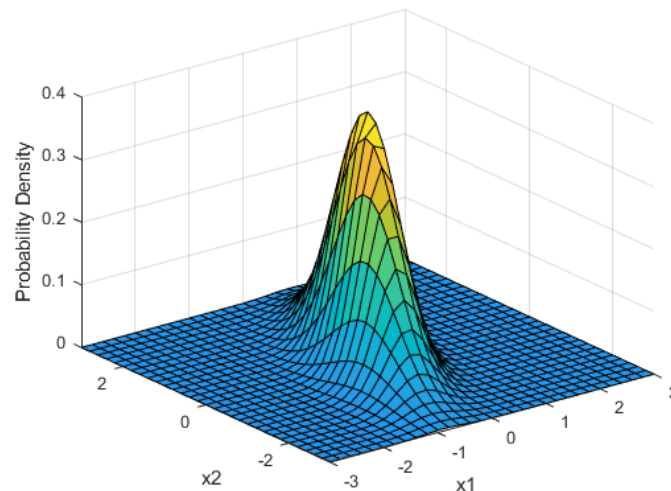
- ✓ Assume that the observed data are drawn from a Gaussian distribution

$X = [x_1, x_2, \dots, x_k] \sim \text{Multivariate (Gaussian) Normal Distribution} (\mu, \Sigma)$

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu) \right]$$

$$\Sigma = \frac{1}{n} \sum_{\mathbf{x}_i \in \mathbf{X}^+} (\mathbf{x}_i - \mu)(\mathbf{x}_i - \mu)^T \text{ (covariance matrix)}$$

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Density Estimation-based Approach

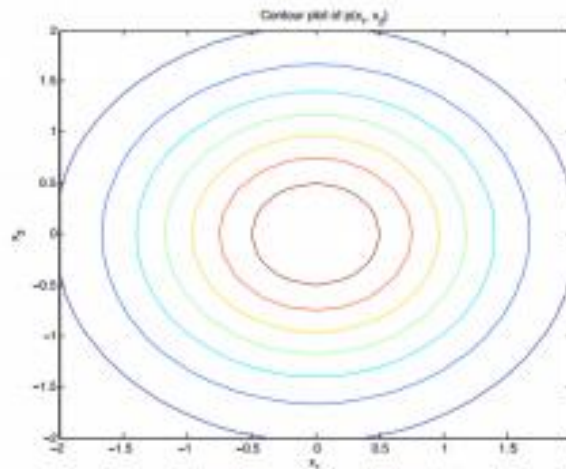
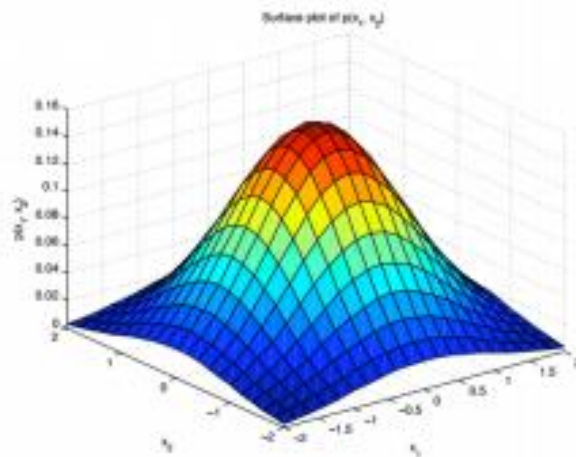
- **Gaussian (Parametric) Density Estimation**

- ✓ Assume that the observed data are drawn from a Gaussian distribution

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left[\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right]$$

Spherical

$$\Sigma = \sigma^2 \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix}$$



(a) Spherical Gaussian (diagonal covariance, equal variances)

Density Estimation-based Approach

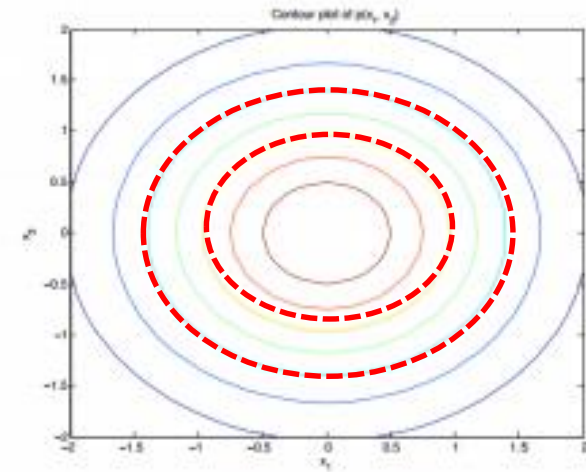
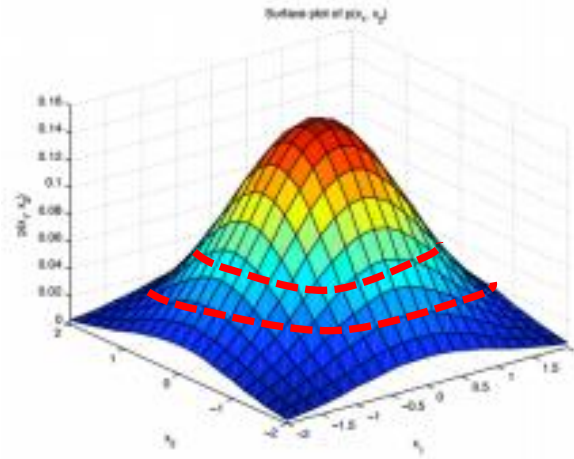
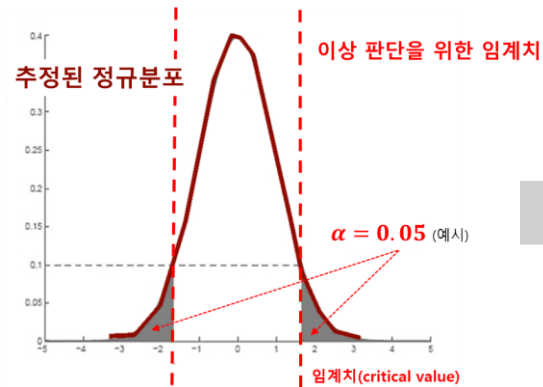
- **Gaussian (Parametric) Density Estimation**

- ✓ Assume that the observed data are drawn from a Gaussian distribution

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left[\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right]$$

Spherical

$$\Sigma = \sigma^2 \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix}$$



(a) Spherical Gaussian (diagonal covariance, equal variances)

Density Estimation-based Approach

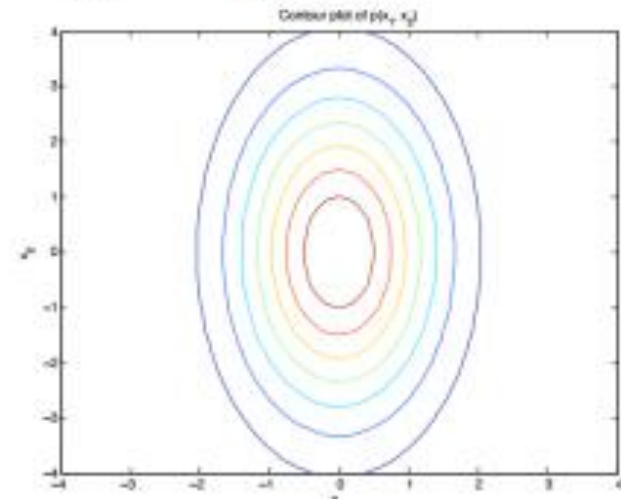
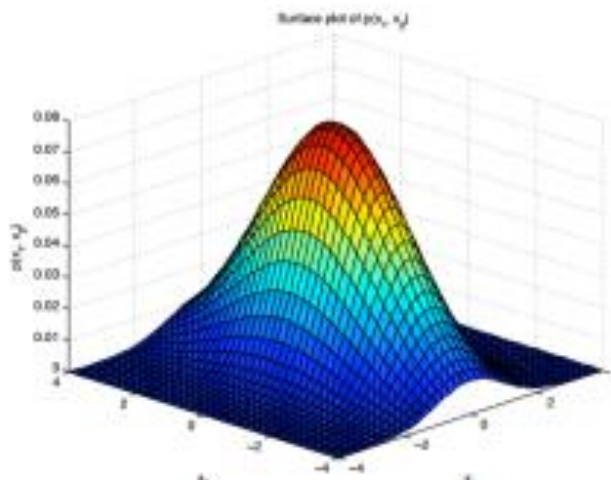
- **Gaussian (Parametric) Density Estimation**

- ✓ Assume that the observed data are drawn from a Gaussian distribution

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left[\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right]$$

Diagonal

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_d^2 \end{bmatrix}$$



Density Estimation-based Approach

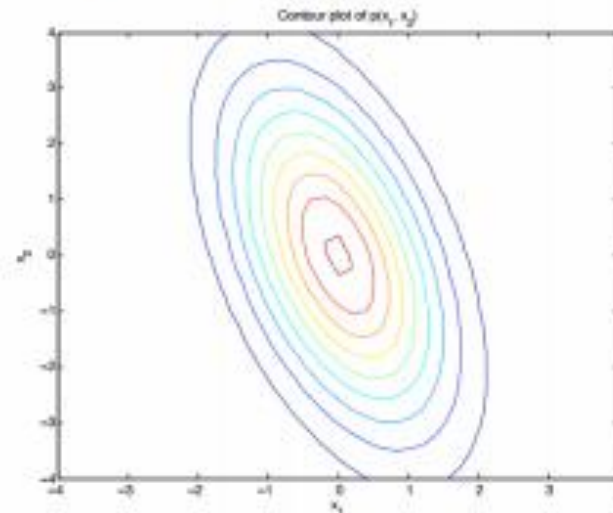
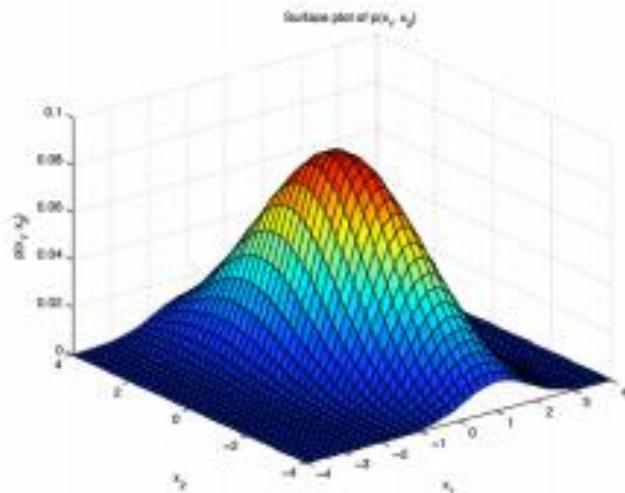
- **Gaussian (Parametric) Density Estimation**

- ✓ Assume that the observed data are drawn from a Gaussian distribution

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left[\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right]$$

Full

$$\Sigma = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1d} \\ \vdots & \ddots & \vdots \\ \sigma_{d1} & \cdots & \sigma_{dd} \end{bmatrix}$$



Density Estimation-based Approach

- **Gaussian (Parametric) Density Estimation**

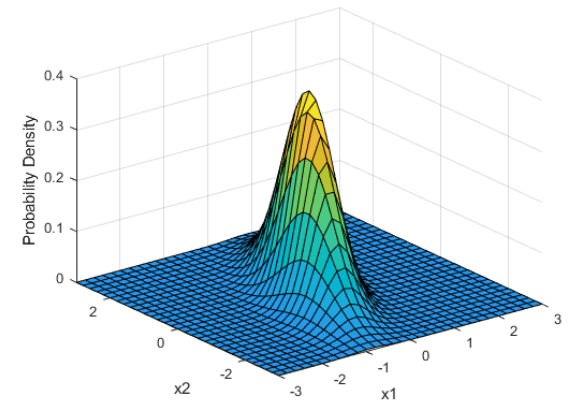
- ✓ Assume that the observed data are drawn from a Gaussian distribution

$P(x)$: 관측치(x)가 정상일 확률

$$P(X) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left[-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)\right],$$

where $\mu = \frac{1}{n} \sum_{\mathbf{x}_i \in \mathbf{X}^+} \mathbf{x}_i$ (mean vector),

$$\Sigma = \frac{1}{n} \sum_{\mathbf{x}_i \in \mathbf{X}^+} (\mathbf{x}_i - \mu)(\mathbf{x}_i - \mu)^T \text{ (covariance matrix)}$$



Density Estimation-based Approach

- Gaussian Density Estimation

- ✓ Advantages

- Insensitive to scaling of the data

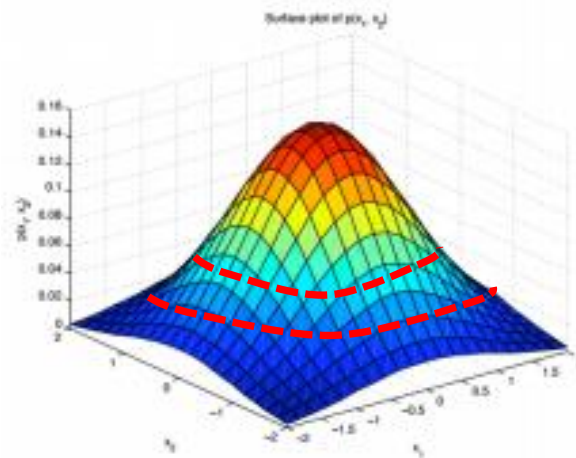
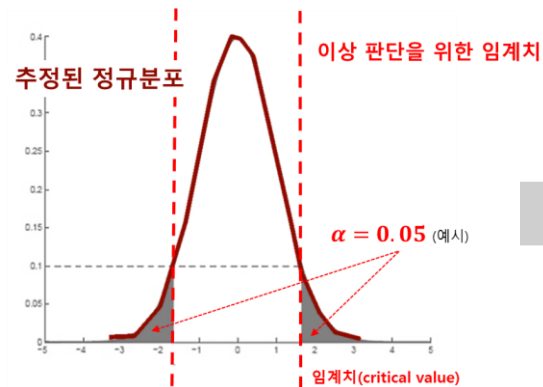
$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left[\frac{1}{2} (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu) \right]$$

- Possible to compute analytically the optimal threshold

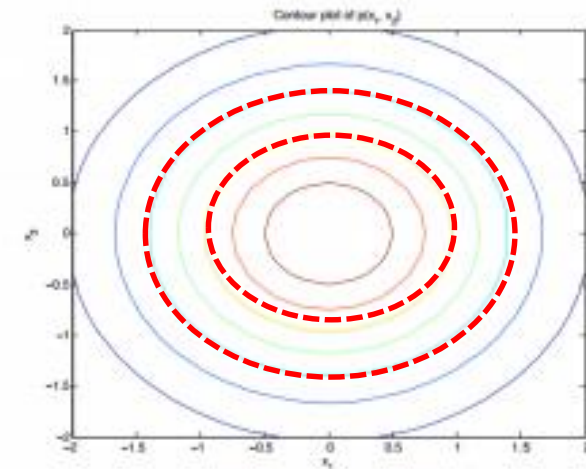
$$\textit{Mahalanobis distance} = (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)$$

Density Estimation-based Approach

Possible to compute analytically optimal threshold



$$\Sigma = \sigma^2 \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix}$$

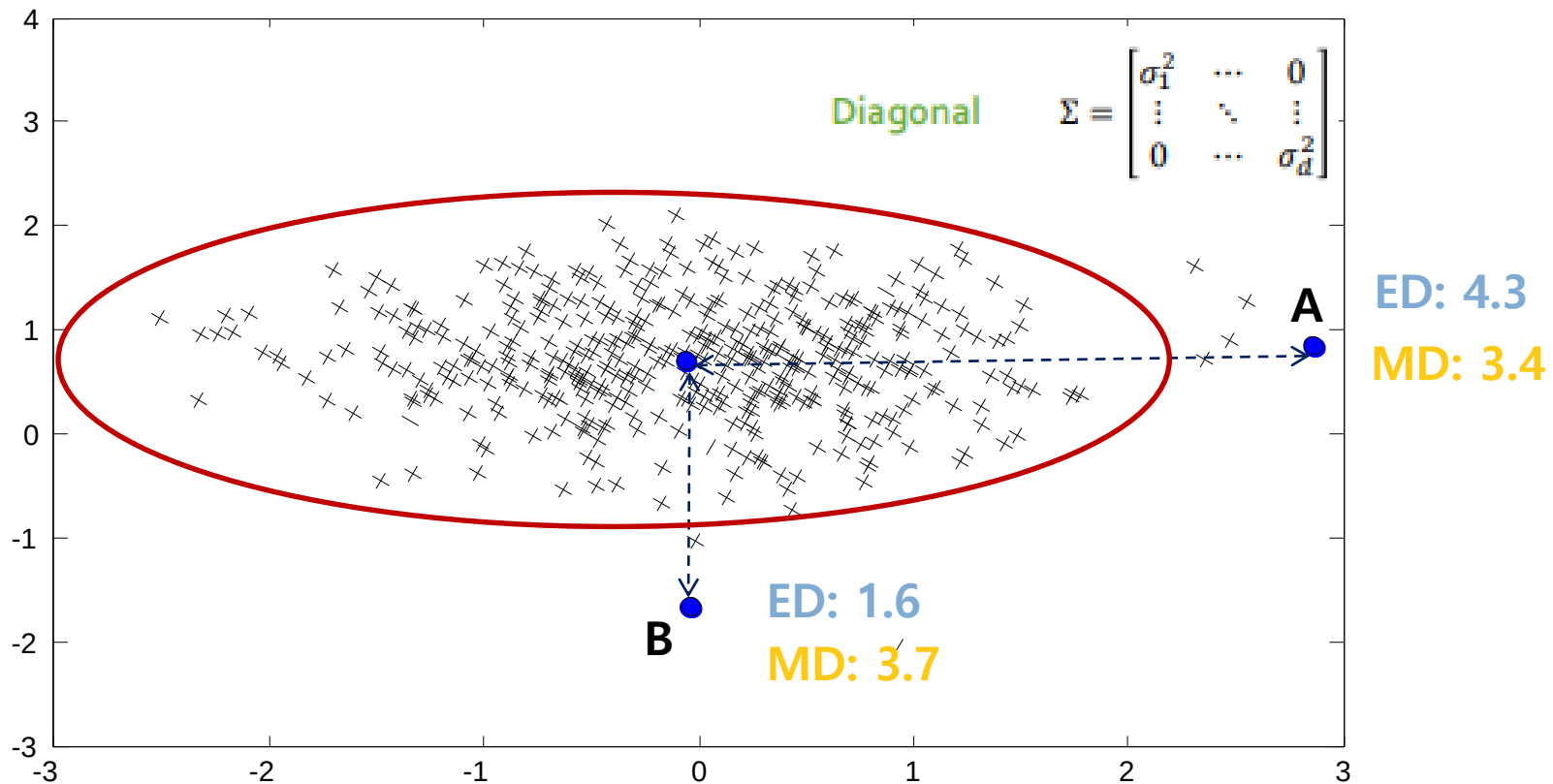


(a) Spherical Gaussian (diagonal covariance, equal variances)

Density Estimation-based Approach

Insensitive to Scaling of the data

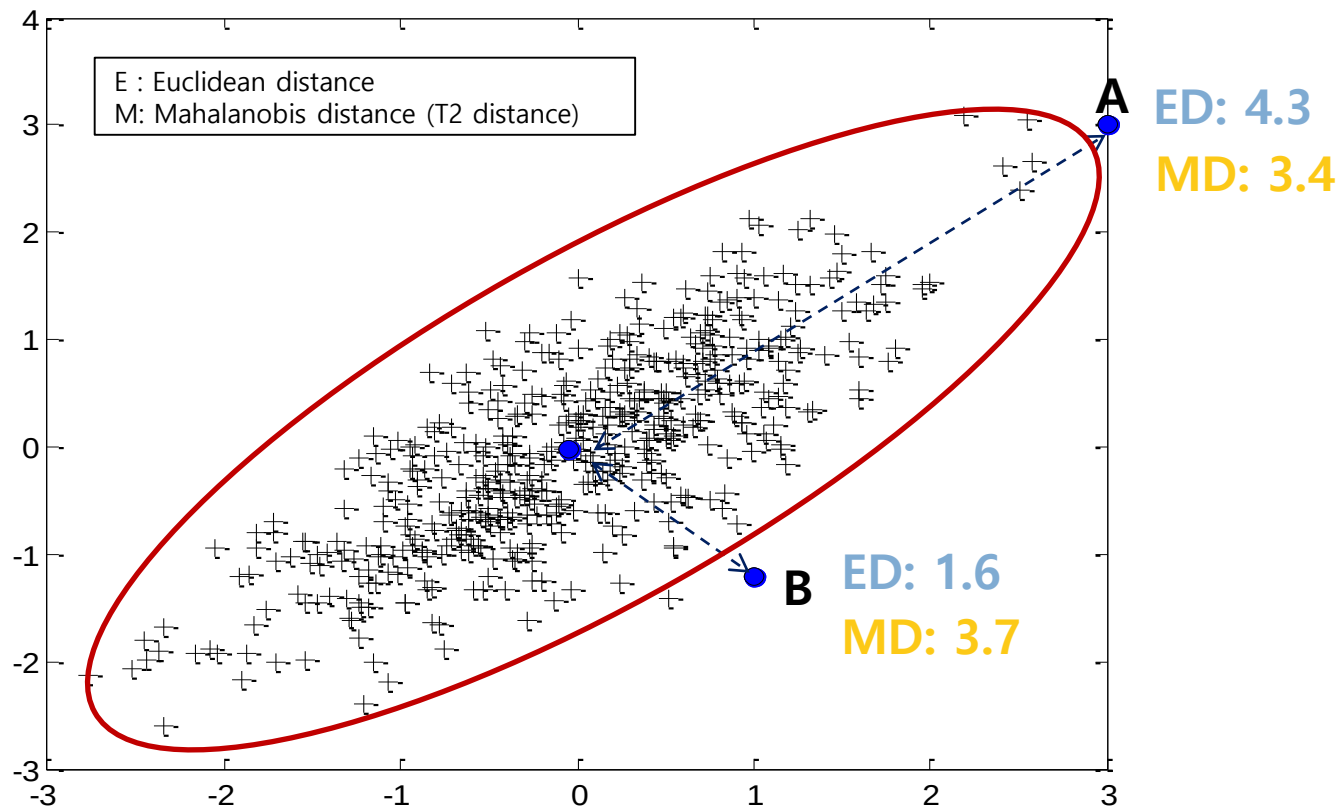
$$\text{Mahalanobis distance} = (x - \mu)^T \Sigma^{-1} (x - \mu)$$



Density Estimation-based Approach

Insensitive to Scaling of the data

$$\text{Mahalanobis distance} = (x - \mu)^T \Sigma^{-1} (x - \mu)$$

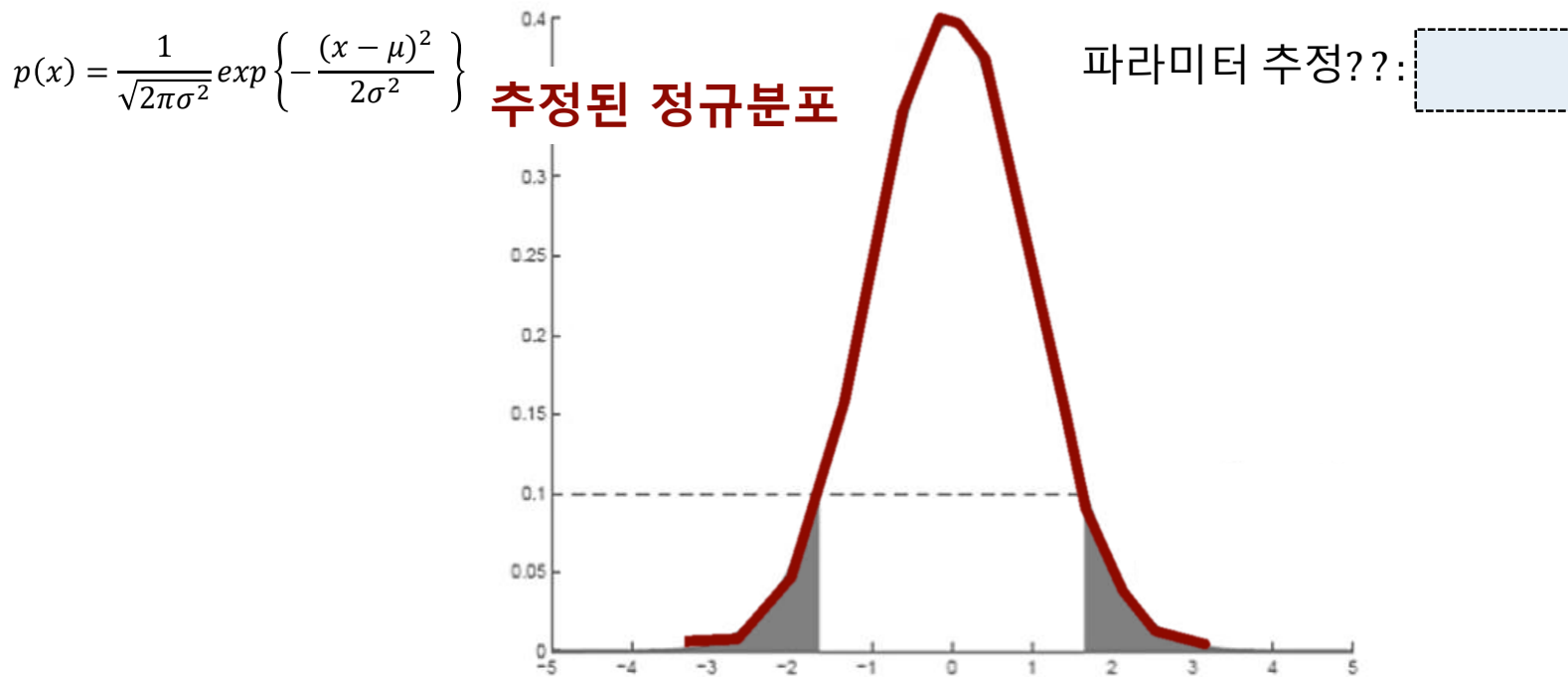


Density Estimation-based Approach

- Gaussian Density Estimation

- ✓ Step 1. 학습모델 구축 I (정규분포 가정)

→ 정상데이터만을 학습데이터로 활용, 정상데이터에 대한 정규분포 추정

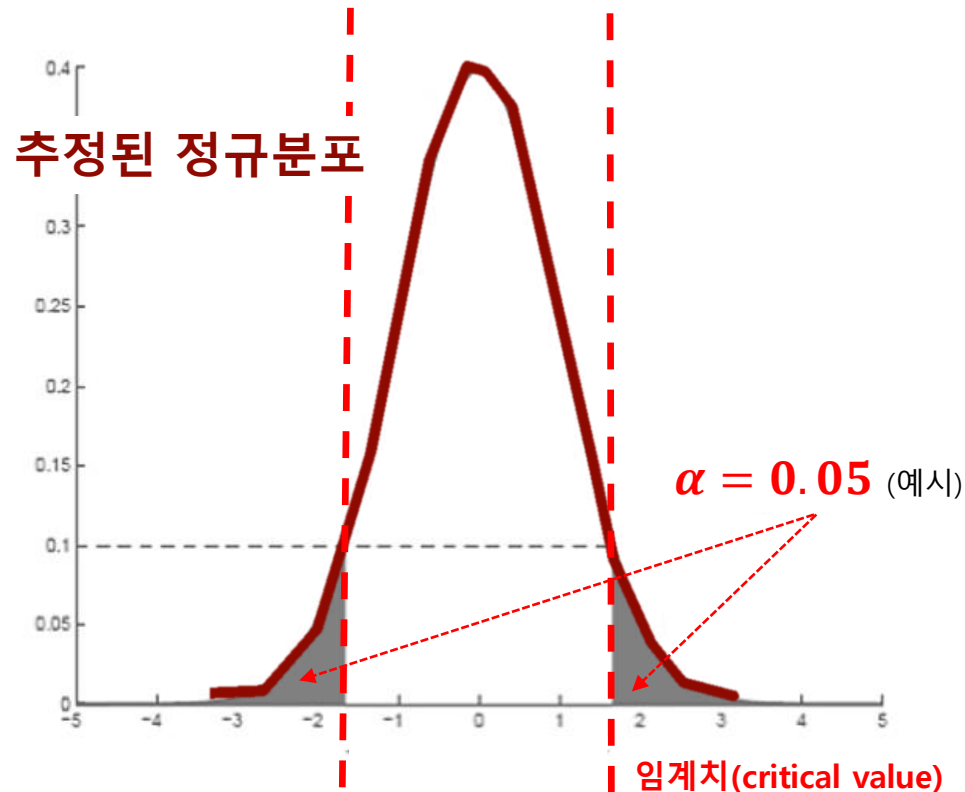


Density Estimation-based Approach

- Gaussian Density Estimation

- ✓ Step 2. Threshold 구축 II (임계치 설정)

→ Anomaly Detection task에 따라, 적정 수준의 임계치를 정하기

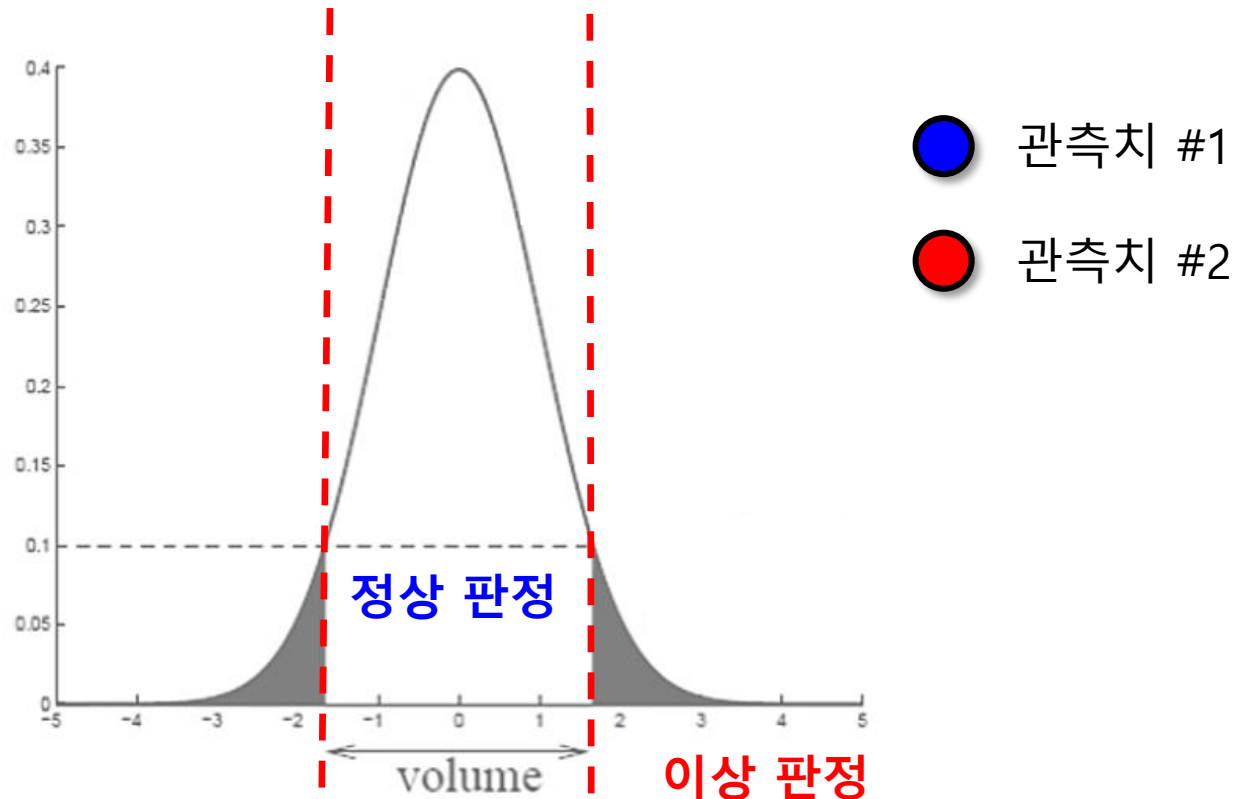


Density Estimation-based Approach

- Gaussian Density Estimation

- ✓ Step 3. 새 관측치에 대해 이상 유무를 판단

- 임계치(with 모델)를 기준으로 새 관측치의 정상/이상 판단



Gaussian Density-based Anomaly Score

- Univariate Gaussian:

$$\text{Anomaly Score} = -\log p(x) = \frac{(x - \mu)^2}{2\sigma^2} + \frac{1}{2} \log(2\pi\sigma^2)$$

- Multivariate Gaussian:

$$e^{-\log(\quad)}$$

$$\text{Anomaly Score} = -\log p(\mathbf{x}) = \frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu) + \frac{1}{2} \log((2\pi)^d |\Sigma|)$$

$$\text{Anomaly score} = \text{Mahalanobis distance} = (\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)$$

1. 해석 용이성:

확률 밀도 $p(x)$ 가 작을수록 더 비정상적인 데이터임을 의미합니다. 하지만 PDF 값은 매우 작은 값으로 나타낼 수 있으므로 로그를 취해 더 직관적인 스코어로 변환합니다.

예: $p(x)=10^{-7} \rightarrow -\log(p(x))=7$ (값이 클수록 더 이상함).

2. 수치적 안정성:

PDF는 지수 함수(exp)를 포함하므로, 계산 중 **언더플로우 문제**가 발생할 수 있습니다. 로그를 취하면 곱셈 연산이 덧셈으로 단순화되어 안정성이 높아집니다.

Decision Boundary

Exact Computation of Decision Boundary

- Univariate Gaussian Probability Density Function

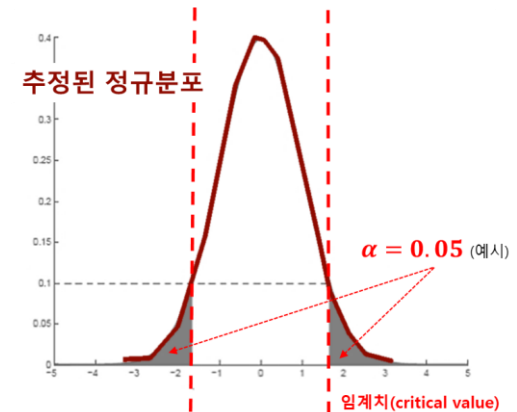
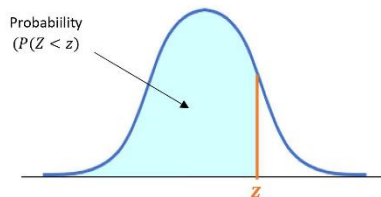
$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}, -\infty < x < \infty$$

- Standard normal distribution pdf

$$Z = \frac{X - \mu}{\sigma}, p(z) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(z)^2}{2\sigma^2}\right\}, -\infty < z < \infty$$

- The cumulative density function (CDF)

$$\Phi(z) = p(Z \leq z) \text{ for } -\infty < z < \infty$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5754
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7258	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
0.7	0.7580	0.7612	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7996	0.8023	0.8051	0.8079	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9430	0.9441
1.6	0.9452	0.9463	0.9474	0.9485	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9615	0.9625	0.9633

Decision Boundary

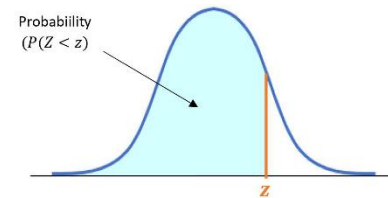
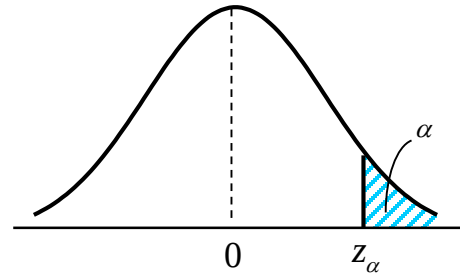
Exact Computation of Decision Boundary

- Univariate Gaussian Probability Density Function

$$\Phi(z) = P(Z > z_\alpha) = \int_{z_\alpha}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \alpha$$

- α = significant level

$$P(Z > z_\alpha) = 1 - P[Z < z_\alpha] = 1 - \Phi(z_\alpha)$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
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0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7258	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
0.7	0.7580	0.7612	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7996	0.8023	0.8051	0.8079	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
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1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9615	0.9625	0.9633

Decision Boundary

Exact Computation of Decision Boundary

- Univariate Gaussian Probability Density Function

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$$

- α = significant level

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = P[Z < z_{\alpha/2}] - P[Z < -z_{\alpha/2}] \quad X = z_{\alpha/2} \sigma + \mu$$

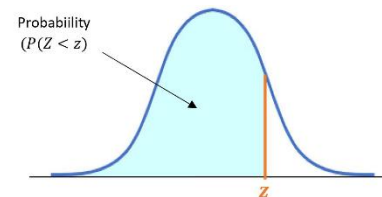
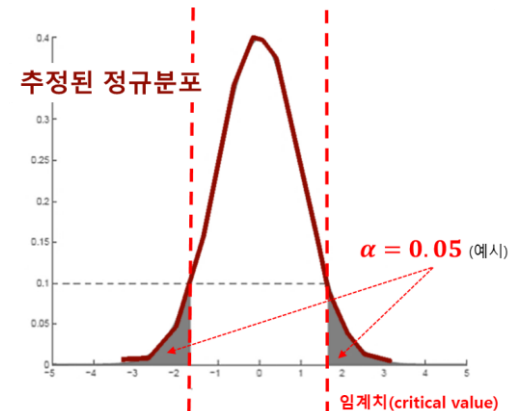
$$P[Z < z_{\alpha/2}] = \Phi(z_{\alpha/2}) = 0.975$$

$$P[Z < z_{\alpha/2}] = \Phi(z_{\alpha/2}) = 0.95$$

$$Z = \frac{X - \mu}{\sigma}$$

- Critical value $x_{upper} = z_{\alpha/2} \sigma + \mu$ and $x_{lower} = -z_{\alpha/2} \sigma + \mu$

- $x_{lower} < x < x_{upper}$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
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0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5754
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0.8	0.7881	0.7910	0.7939	0.7967	0.7996	0.8023	0.8051	0.8079	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9430	0.9441
1.6	0.9452	0.9463	0.9474	0.9485	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633

Decision Boundary

Exact Computation of Decision Boundary

- Univariate Gaussian Probability Density Function

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$$

- α = significant level

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = P[Z < z_{\alpha/2}] - P[Z < -z_{\alpha/2}]$$

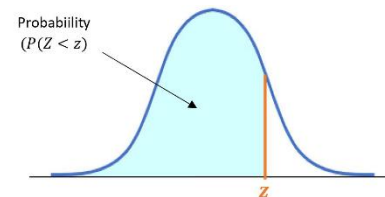
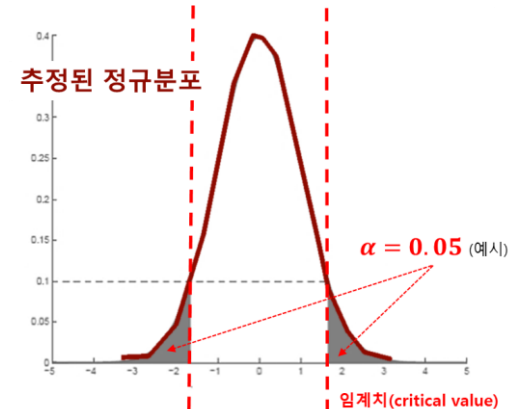
$$P[Z < z_{\alpha/2}] = \Phi(z_{\alpha/2}) = 0.975$$

$$P[Z < z_{\alpha/2}] = \Phi(z_{\alpha/2}) = 0.95$$

$$Z = \frac{X - \mu}{\sigma} \quad x = z_{\alpha/2} \sigma + \mu$$

- Critical value $x_{\text{upper}} = z_{\alpha/2} \sigma + \mu$ and $x_{\text{lower}} = -z_{\alpha/2} \sigma + \mu$

- Critical value = $-\log(p(x_{\text{upper}}))$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5754
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7258	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
0.7	0.7580	0.7612	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7996	0.8023	0.8051	0.8079	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
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1.6	0.9452	0.9463	0.9474	0.9485	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633

Decision Boundary

Approximate Computation of Decision Boundary

- Univariate Gaussian:

$$\text{Anomaly Score} = -\log p(x) = \frac{(x - \mu)^2}{2\sigma^2} + \frac{1}{2} \log(2\pi\sigma^2)$$

경험적 백분위수 (Empirical Percentiles)

- 정상 데이터에서 특정 백분위수를 기준으로 임계치를 설정합니다.
- 예를 들어, 상위 95% 백분위수를 기준으로 하면 하위 5% 데이터를 이상치로 간주할 수 있습니다.

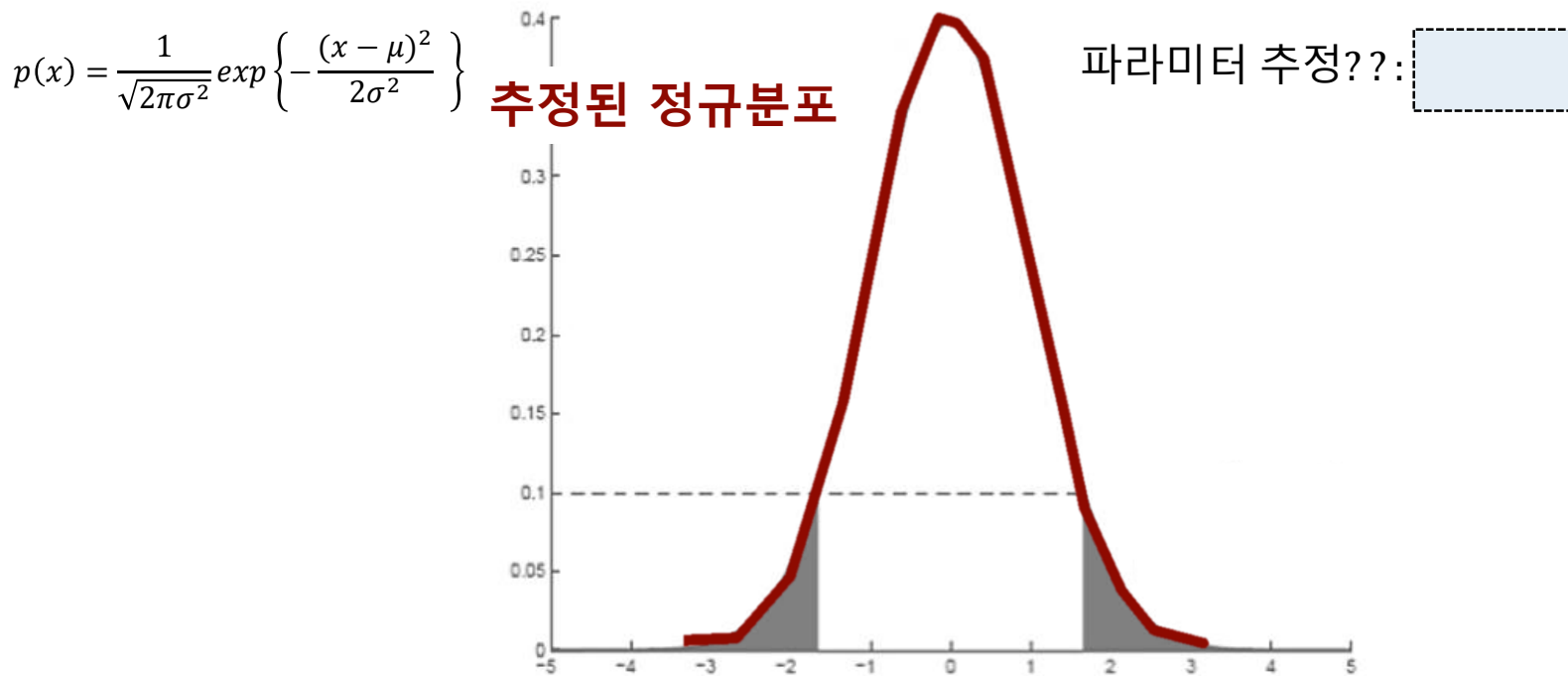
anomaly score ?

Density Estimation-based Approach

- Gaussian Density Estimation

- ✓ Step 1. 학습모델 구축 I (정규분포 가정)

→ 정상데이터만을 학습데이터로 활용, 정상데이터에 대한 정규분포 추정

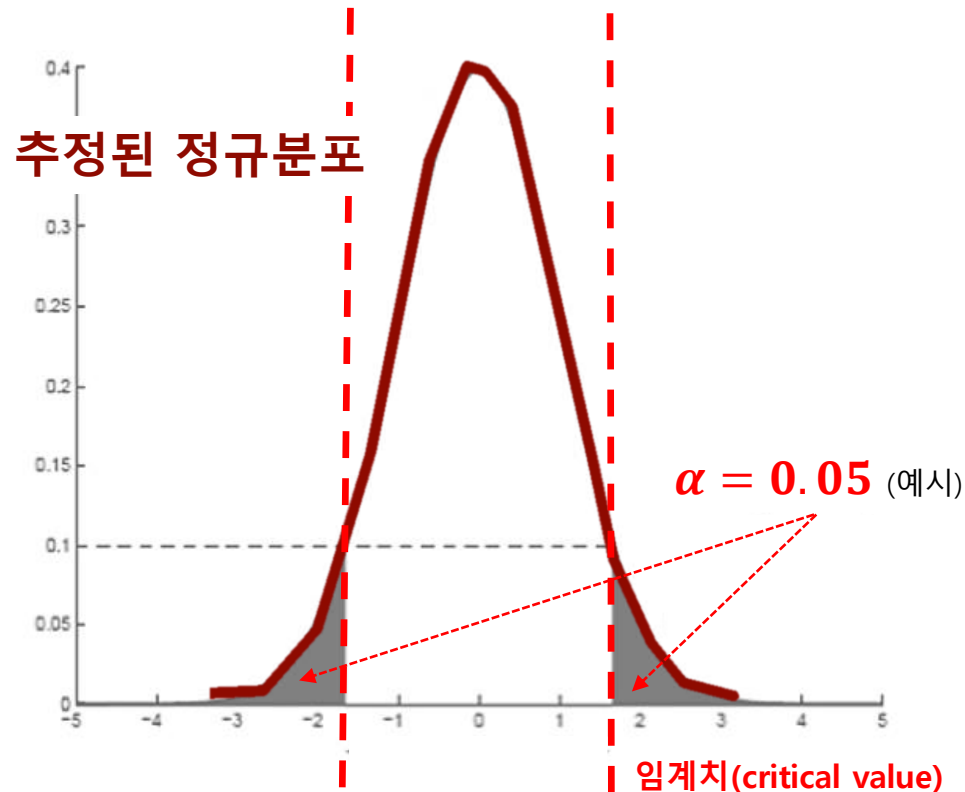


Density Estimation-based Approach

- Gaussian Density Estimation

- ✓ Step 2. Threshold 구축 II (임계치 설정)

→ Anomaly Detection task에 따라, 적정 수준의 임계치를 정하기

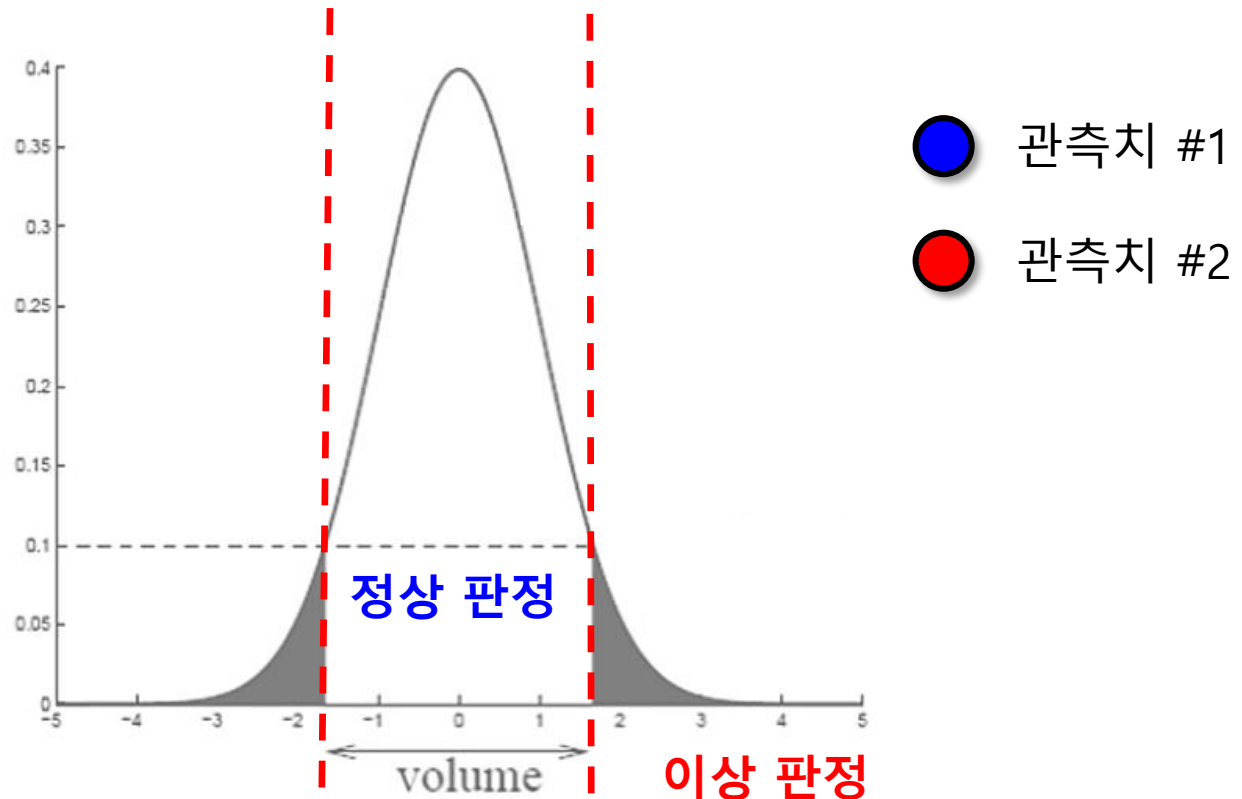


Density Estimation-based Approach

- Gaussian Density Estimation

- ✓ Step 3. 새 관측치에 대해 이상 유무를 판단

- 임계치(with 모델)를 기준으로 새 관측치의 정상/이상 판단



Decision Boundary

- **Multivariate Gaussian Probability Density Function**

Mahalanobis 거리 기반 임계치:

- Mahalanobis 거리 D_M^2 는 다음과 같이 정의됩니다:

$$D_M^2 = (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)$$

- 이 거리는 자유도 d 를 가진 카이제곱 분포(χ_d^2)를 따릅니다.
- 유의 수준 α 에 해당하는 카이제곱 임계값 T'_α 는 다음과 같이 계산됩니다:

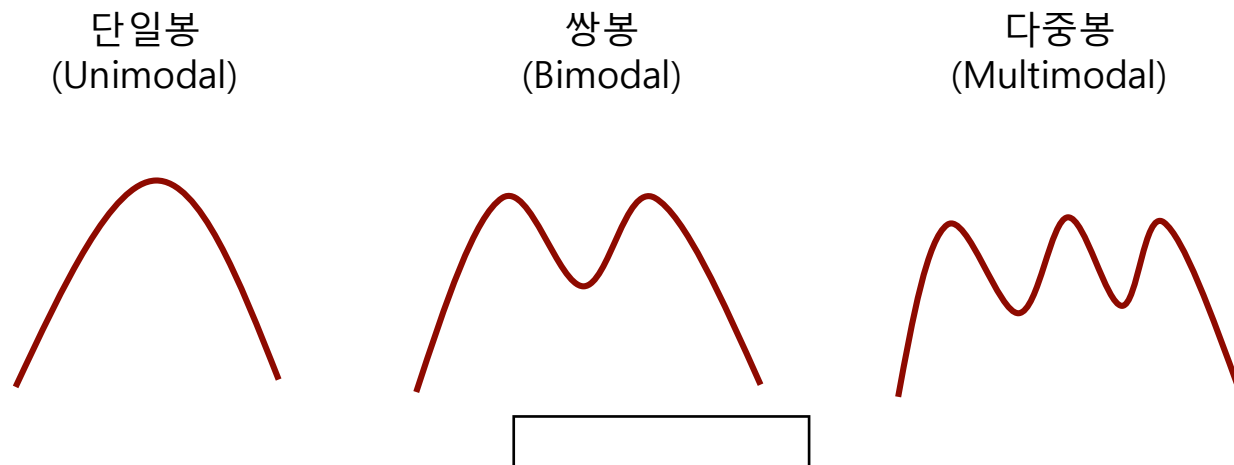
$$P(D_M^2 > T'_\alpha) = \alpha$$

경험적 백분위수 (Empirical Percentiles)

- 정상 데이터에서 특정 백분위수를 기준으로 임계치를 설정합니다.
- 예를 들어, 상위 95% 백분위수를 기준으로 하면 하위 5% 데이터를 이상치로 간주할 수 있습니다.

Density Estimation-based Approach

- 혼합가우시안밀도추정 (Mixture of Gaussian Density Estimation)
 - ✓ 정규분포는 unimodal를 가정
→ Model bias가 높음
 - ✓ MoG (Mixture of Gaussian)는 Gaussian Density 방식의 확장방식
MOG = Gaussian Mixture Model (GMM) 동일 용어

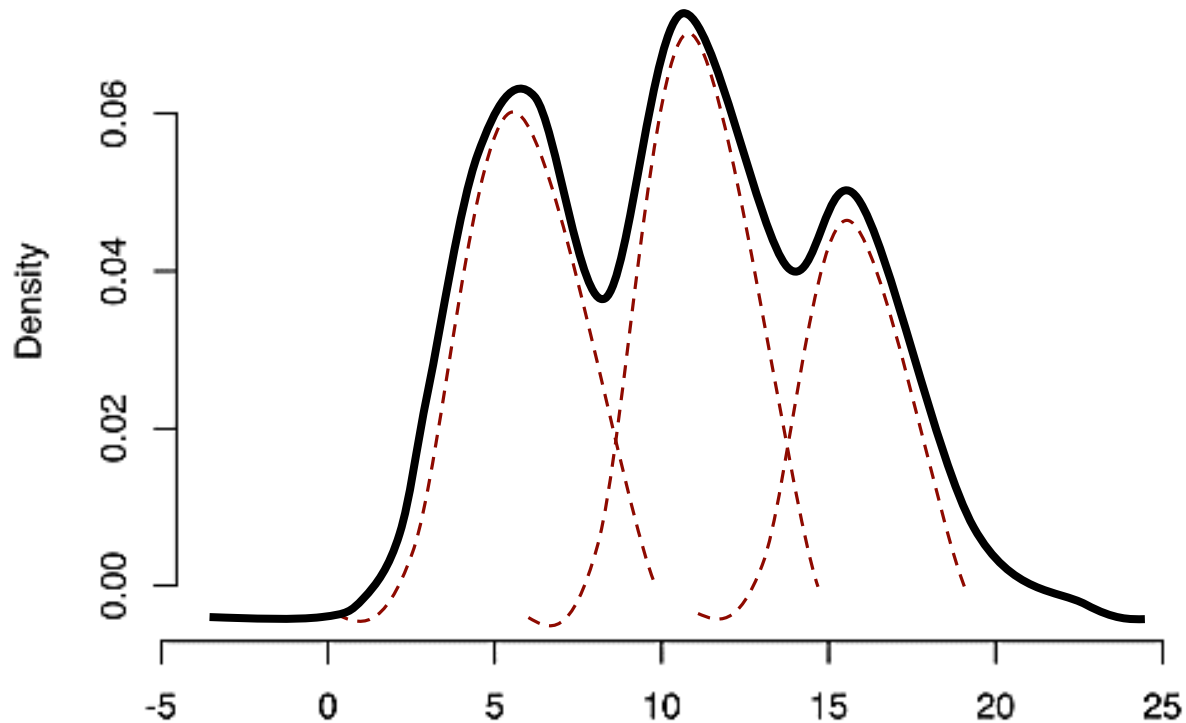


Density Estimation-based Approach

- 가우시안혼합모델 (Gaussian Mixture Model)

- ✓ 봉이 3개 짜리 밀도를 갖고 있다면?

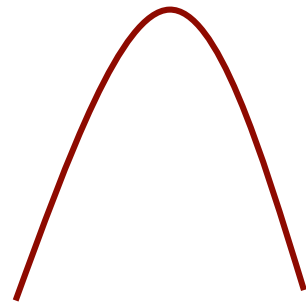
- 3개의 가우시안밀도추정 모델을 만들어 적합(fitting)



Density Estimation-based Approach

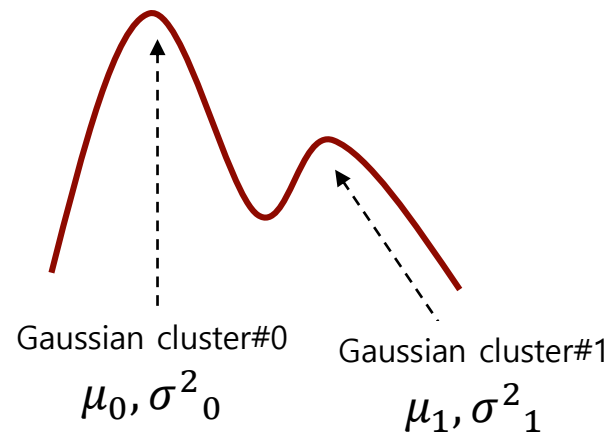
- 가우시안혼합모델 (Gaussian Mixture Model)
 - ✓ MoG는 정규밀도추정의 단순한 선형 합(linear combination)

단일봉
(Unimodal)

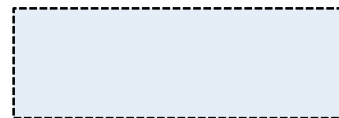
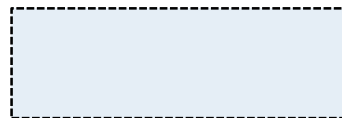


μ, σ^2

쌍봉
(Bimodal)



파라미터 비교

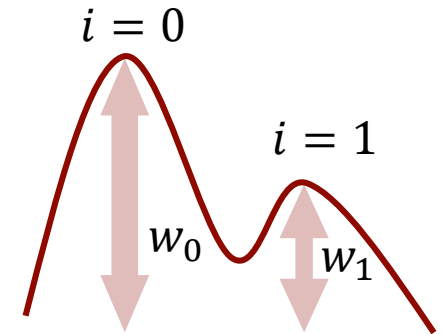


Density Estimation-based Approach

- 가우시안혼합모델 (Gaussian Mixture Model)

- ✓ MoG는 정규밀도추정의 단순한 선형 합(linear combination)
- ✓ Mean μ_i , Variance σ^2_i , Size w_i
- ✓ 확률분포 (Probability distribution):

$$P(x) = \sum_{i=1}^T w_i \cdot g(x|\mu_i, \Sigma_i)$$



- ✓ Equivalent "latent variable" form for z :

$$P(z = i) = w_i \quad \text{각 Gaussian cluster의 확률}$$

$$P(x | z = i) = g(x | \mu_i, \Sigma_i) \quad \text{관측치 } x \text{가 특정 Gaussian cluster에 속할 확률}$$

Density Estimation-based Approach

- 가우시안혼합모델 (Gaussian Mixture Model)

- ✓ MoG는 정규밀도추정의 단순한 선형 합(linear combination)

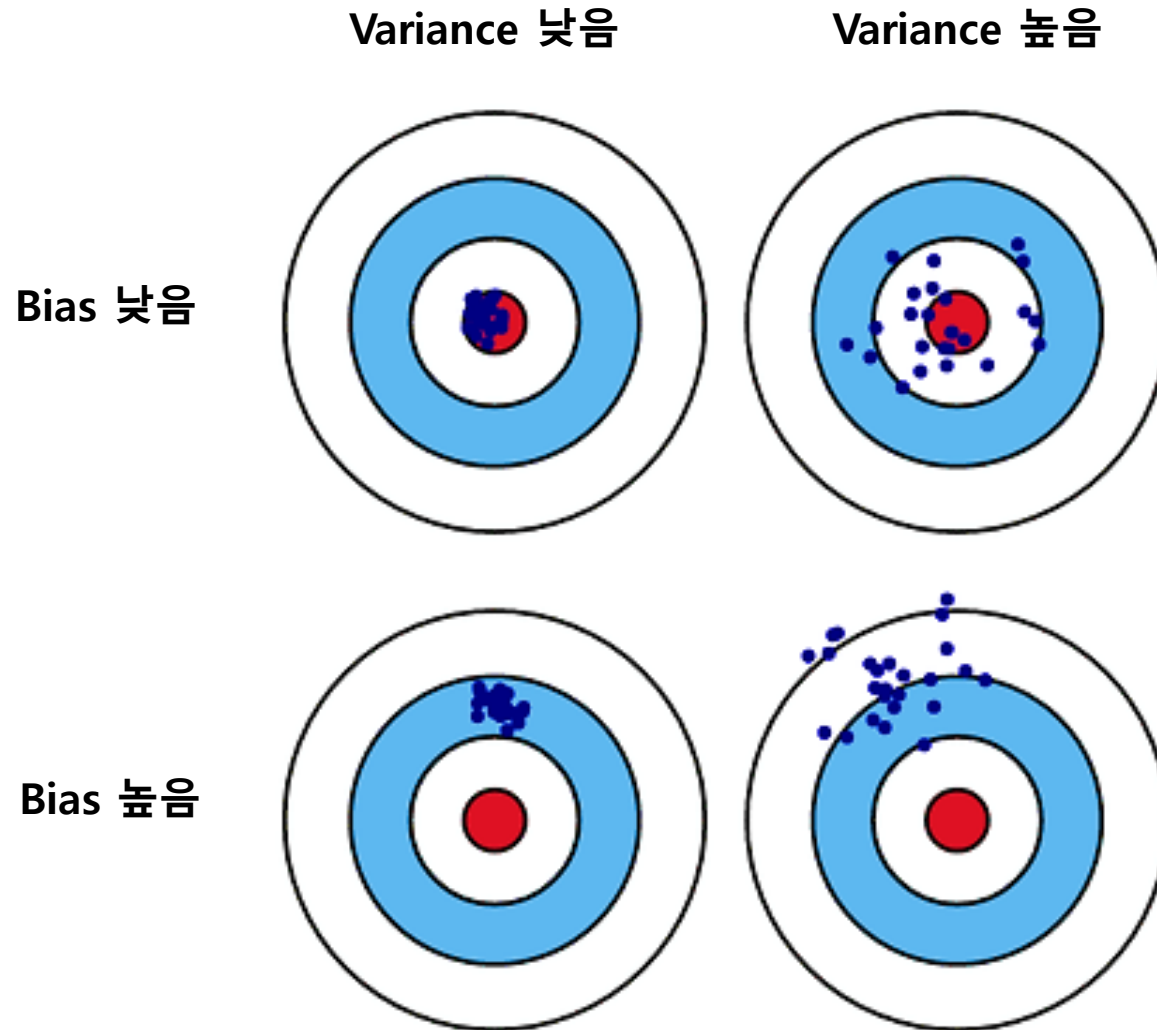
$$P(x | \lambda) = \sum_{i=1}^T w_i \cdot g(x | \mu_i, \Sigma_i)$$

$$g(x | \mu_i, \Sigma_i) = \frac{1}{(2\pi)^{D/2} |\Sigma_i|^{1/2}} \exp \left[-\frac{1}{2} (x - \mu_i)^T \Sigma_i^{-1} (x - \mu_i) \right]$$

$$\lambda = \{w_i, \mu_i, \Sigma_i\} \quad i = 1, \dots, M$$

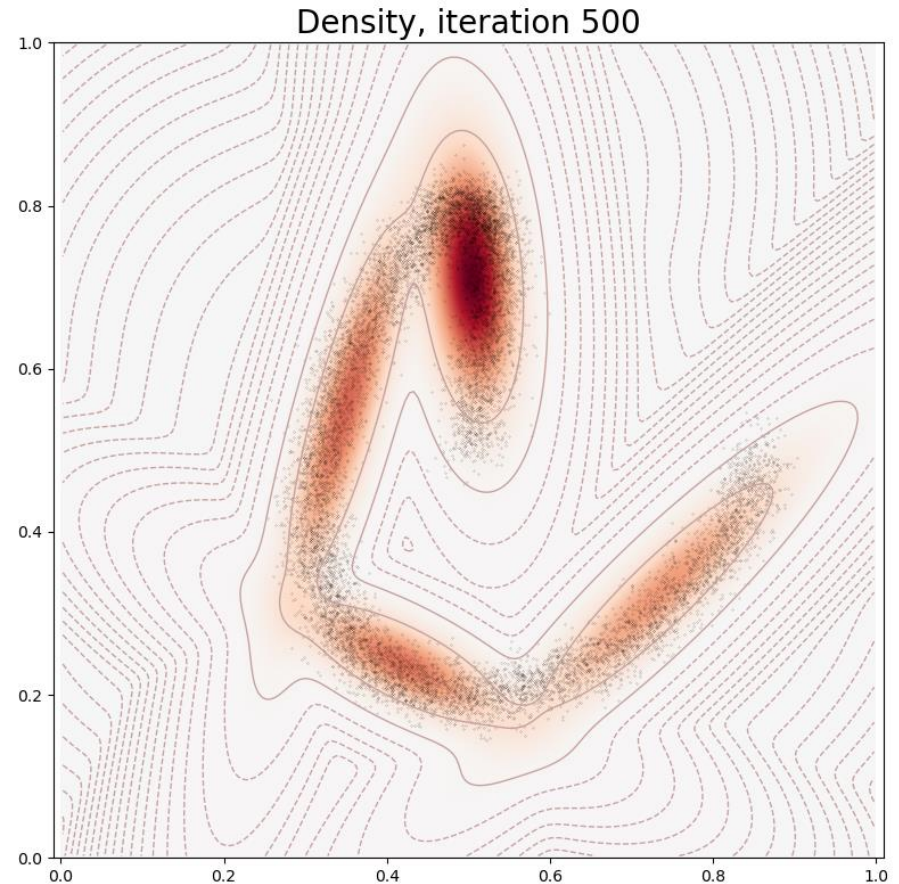
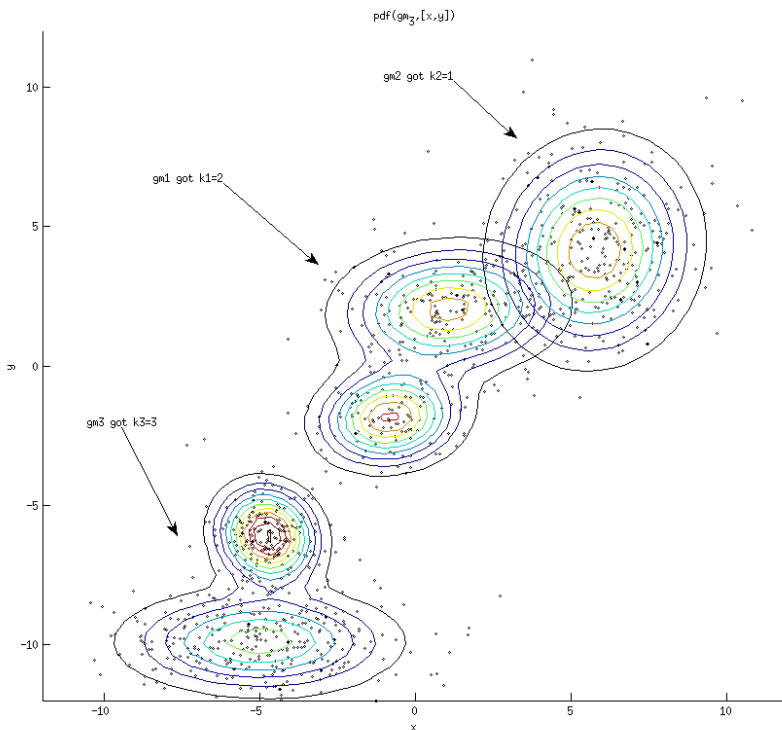
- ✓ Bias 면에서 혼합가우시안밀도추정 < 가우시안밀도추정

Bias 와 Variance의 개념 예시



혼합가우시안밀도추정 vs. 가우시안밀도추정

- 가우시안혼합모델 (Gaussian Mixture Model)
 - ✓ MoG는 정규밀도추정의 단순한 선형 합(linear combination)



GMM의 파라미터 추정

- 가우시안혼합모델 (Gaussian Mixture Model)

- ✓ 파라미터의 추정 알고리즘

- Expectation-Maximization algorithm (EM algorithm)

- ✓ **Expectation**

$$\Pr(i|x_t, \lambda) = \frac{w_i g(x|\mu_i, \Sigma_i)}{\sum_{k=1}^M w_k g(x|\mu_k, \Sigma_k)}$$

Compute $\Pr(i|x_t, \lambda)$
Probability that responsibility of
cluster i given x

- ✓ **Maximization**

Mixture weight: $w_i = \frac{1}{T} \sum_{t=1}^T \Pr(i|x_t, \lambda)$

Means : $u_i = \frac{\sum_{t=1}^T \Pr(i|x_t, \lambda) x_t}{\sum_{t=1}^T \Pr(i|x_t, \lambda)}$

coariances:

$$\Sigma_i = \frac{\sum_{t=1}^T \Pr(i|x_t, \lambda)}{\sum_{t=1}^T \Pr(i|x_t, \lambda)} (x_t - u_i)(x_t - u_i)^T$$

GMM의 파라미터 추정

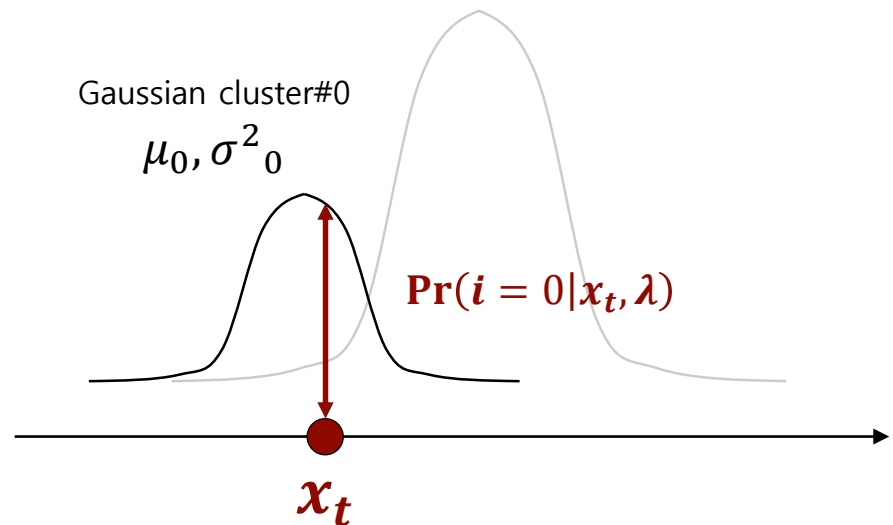
- 가우시안혼합모델 (Gaussian Mixture Model)

- ✓ 파라미터의 추정 알고리즘

- Expectation-Maximization algorithm (EM algorithm)

- ✓ Expectation

$$\Pr(i|x_t, \lambda) = \frac{w_i g(x_t|\mu_i, \Sigma_i)}{\sum_{k=1}^M w_k g(x_t|\mu_k, \Sigma_k)}$$



GMM의 파라미터 추정

- 가우시안혼합모델 (Gaussian Mixture Model)

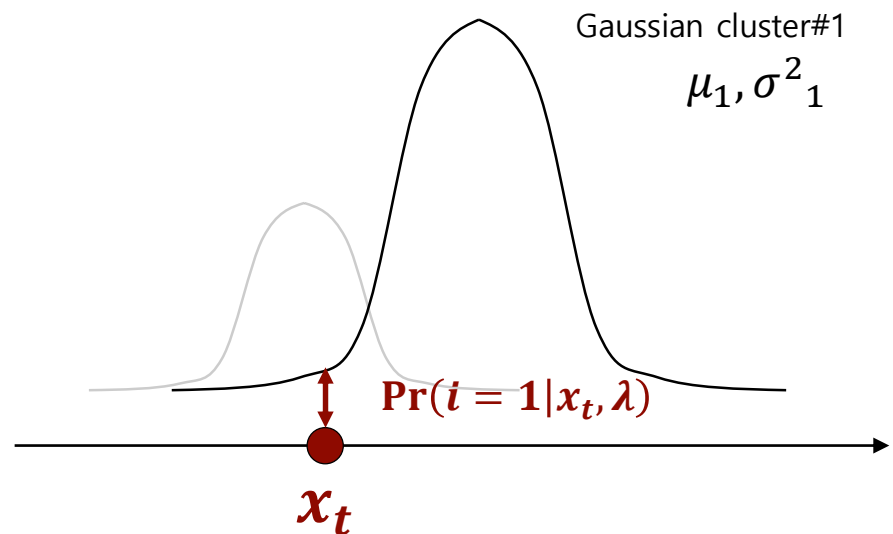
- ✓ 파라미터의 추정 알고리즘

- Expectation-Maximization algorithm (EM algorithm)

- ✓ Expectation

$$\Pr(i|x_t, \lambda) = \frac{w_i g(x_t|\mu_i, \Sigma_i)}{\sum_{k=1}^M w_k g(x_t|\mu_k, \Sigma_k)}$$

관측치 x_t 가 개별 cluster에
속할 확률 계산 후
 i^{th} cluster에 할당



GMM의 파라미터 추정

- 가우시안혼합모델 (Gaussian Mixture Model)

- ✓ 파라미터의 추정 알고리즘

- Expectation-Maximization algorithm (EM algorithm)

- ✓ Maximization

x i 개

- i^{th} cluster의 확률
- i^{th} cluster의 평균
- i^{th} cluster의 분산

Mixture weight: $w_i = \frac{1}{T} \sum_{t=1}^T \Pr(i|x_t, \lambda)$

Means : $u_i = \frac{\sum_{t=1}^T \Pr(i|x_t, \lambda) x_t}{\sum_{t=1}^T \Pr(i|x_t, \lambda)}$

Covariances:

$$\Sigma_i = \frac{\sum_{t=1}^T \Pr(i|x_t, \lambda)}{\sum_{t=1}^T \Pr(i|x_t, \lambda)} (x_t - u_i)(x_t - u_i)^T$$

GMM의 파라미터 추정

Algorithm 1: EM algorithm for GMM

Input : a given data $X = \{x_1, x_2, \dots, x_n\}$

Output: $\pi = \{\pi_1, \pi_2, \dots, \pi_K\}$,

$\mu = \{\mu_1, \mu_2, \dots, \mu_K\}$,

$\Sigma = \{\Sigma_1, \Sigma_2, \dots, \Sigma_K\}$

1 Randomly initialize π, μ, Σ

2 **for** $t = 1 : T$ **do**

3 // E-step

4 **for** $n = 1 : N$ **do**

5 **for** $k = 1 : K$ **do**

6 $\gamma(z_{nk}) = \frac{\pi_k N(x_n | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j N(x_n | \mu_j, \Sigma_j)}$

7 **end**

8 **end**

9 // M-step

10 **for** $k = 1 : K$ **do**

11 $\mu_k = \frac{\sum_{n=1}^N \gamma(z_{nk}) x_n}{\sum_{n=1}^N \gamma(z_{nk})}$

12 $\Sigma_k = \frac{\sum_{n=1}^N \gamma(z_{nk}) (x_n - \mu_k)(x_n - \mu_k)^T}{\sum_{n=1}^N \gamma(z_{nk})}$

13 $\pi_k = \frac{1}{N} \sum_{n=1}^N \gamma(z_{nk})$

14 **end**

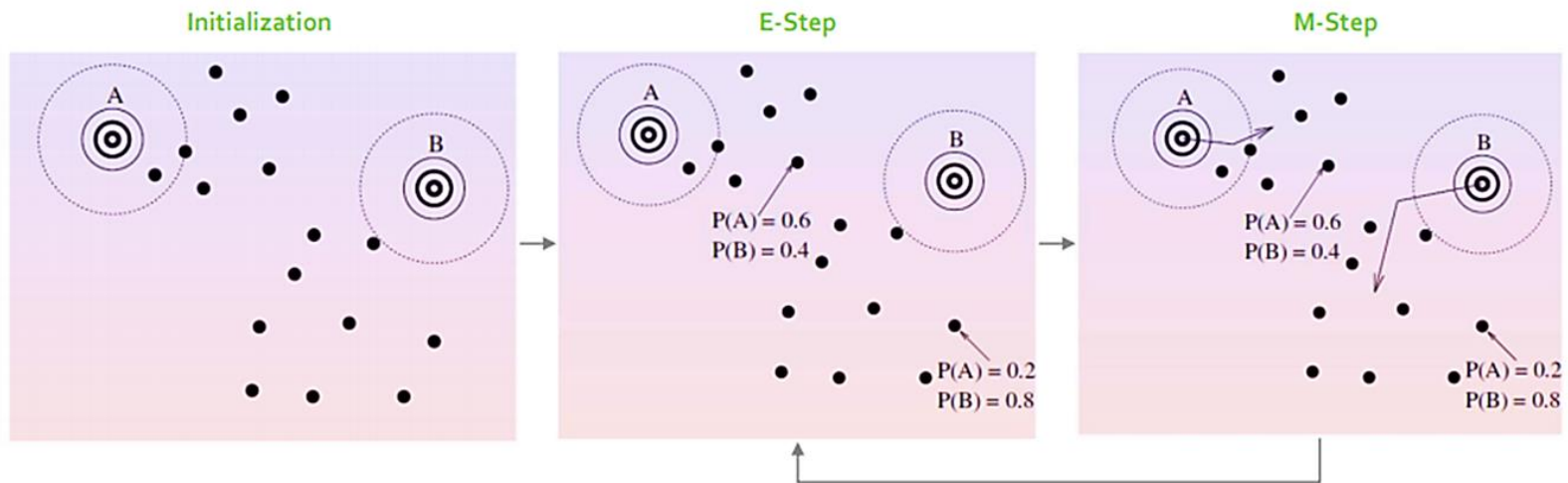
15 **end**

GMM의 파라미터 추정

- 가우시안혼합모델 (Gaussian Mixture Model)

- ✓ 파라미터의 추정 알고리즘

- Expectation-Maximization algorithm (EM algorithm)



- ✓ E-Step: 주어진 파라미터를 기준으로 조건부확률 계산

- ✓ M-Step: 우도(likelihood)를 최대화 하도록 파라미터 수정

GMM의 파라미터 추정

- 가우시안혼합모델 (Gaussian Mixture Model)

- ✓ 파라미터의 추정 알고리즘

- Expectation-Maximization algorithm (EM algorithm)

$$\log P(x) = \sum_t \log \left[\sum_i w_i \cdot g(x_t | \mu_i, \Sigma_i) \right]$$

- ✓ E step – M step 반복 수행 (수렴될 때까지)

- 수렴이 보장되어 있음 (Convergence guaranteed)

- 단, 전역 최적해가 보장되지는 않음 (Local optimal solution)

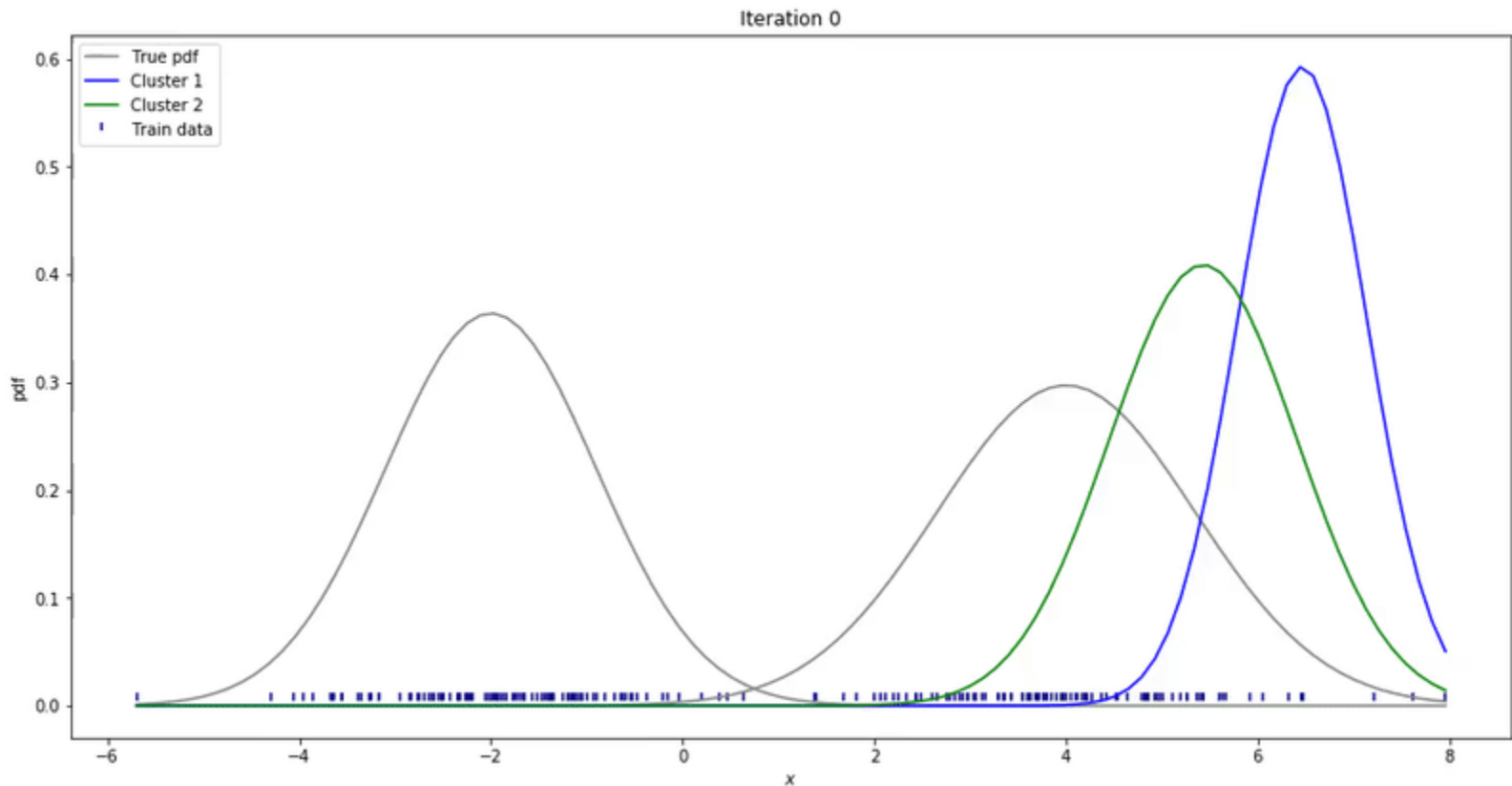
- ※ 초기해 및 몇 개의 cluster로 설정할지가 중요함

가우시안혼합모델 파라미터 탐색 과정

- 가우시안혼합모델 (Gaussian Mixture Model)

- ✓ 파라미터의 추정

- 적합 되어가는 과정 가시화 (Iteration ▲)



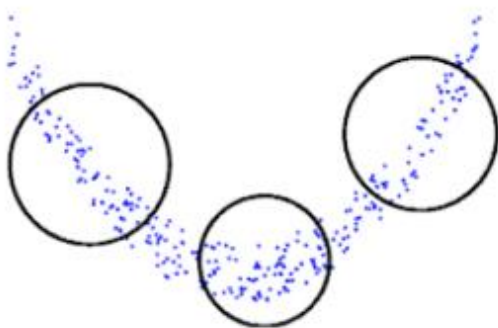
공분산 형태 정의에 따른 학습속도 개선

- 가우시안혼합모델 (Gaussian Mixture Model)

- ✓ 각 Gaussian cluster의 공분산(covariance)를 다 구하는게 어려우니 계산량 줄이기 위한 방법

Spherical

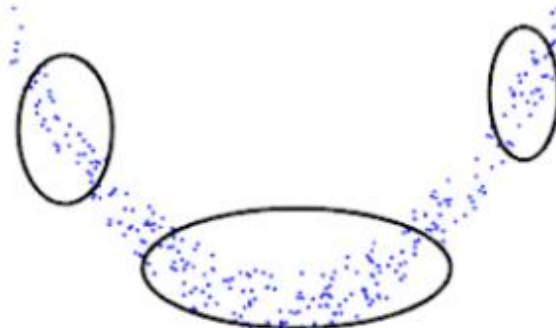
$$\Sigma_j = \text{diag}(\sigma_j^2, \sigma_j^2, \dots, \sigma_j^2) = \sigma_j^2 I$$



- Less precise
- Very efficient to compute

Diagonal

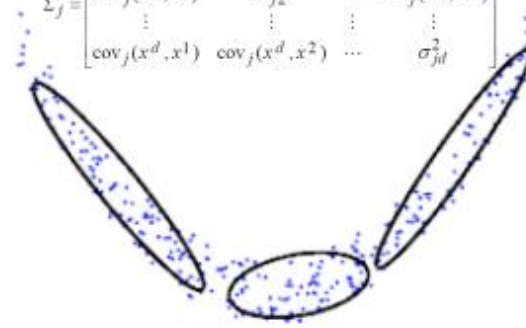
$$\Sigma_j = \text{diag}(\sigma_{j1}^2, \sigma_{j2}^2, \dots, \sigma_{jd}^2)$$



- More precise
- Efficient to compute

Full

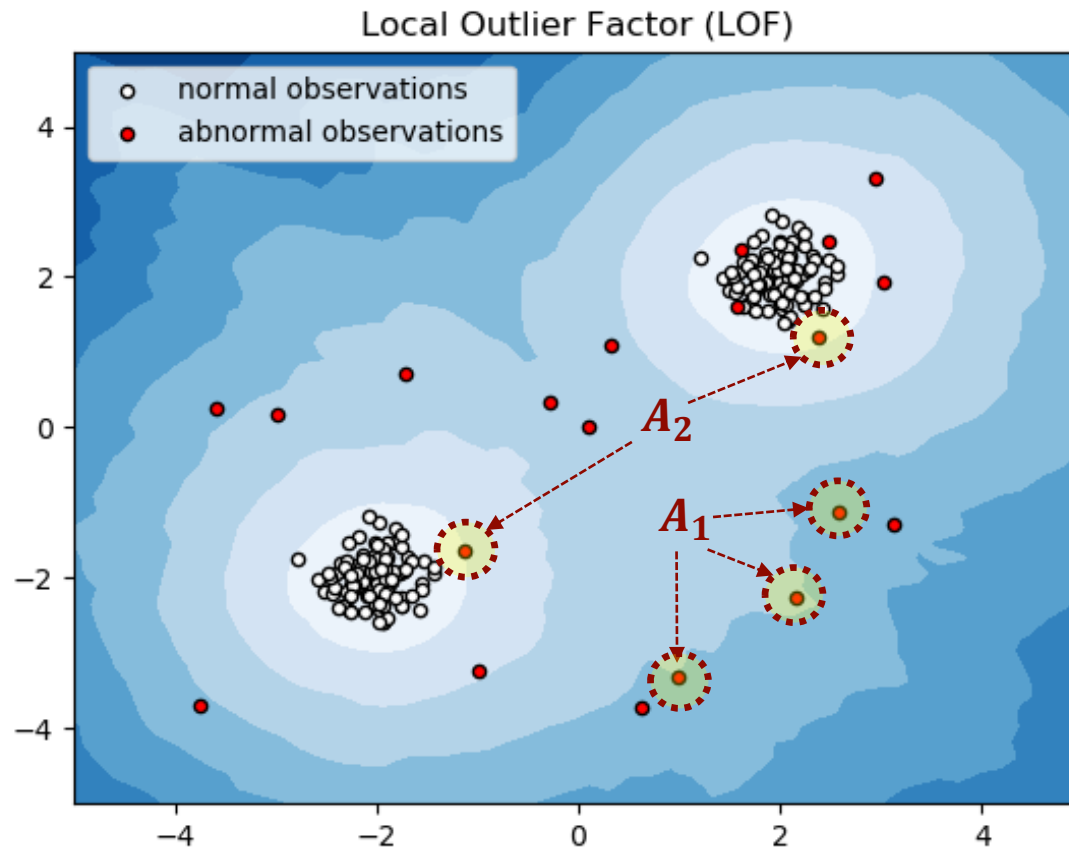
$$\Sigma_j = \begin{bmatrix} \sigma_{j1}^2 & \text{cov}_j(x^1, x^2) & \dots & \text{cov}_j(x^1, x^d) \\ \text{cov}_j(x^2, x^1) & \sigma_{j2}^2 & \dots & \text{cov}_j(x^2, x^d) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}_j(x^d, x^1) & \text{cov}_j(x^d, x^2) & \dots & \sigma_{jd}^2 \end{bmatrix}$$



- Very precise
- Less efficient to compute

Local outlier factor (LOF)

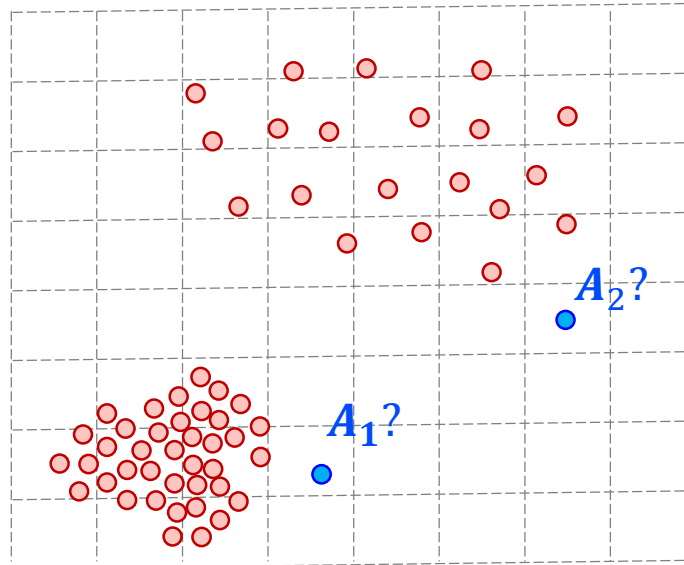
- Local outlier factor (LOF): 지역적 특징(locality pattern) 중에서도 밀도(density)에 대한 고려하여 이상치(outlier)의 점수화를 방안 부재



Local outlier factor (LOF)

- Local outlier factor (LOF): 특정 관측치의 이웃점들을 지역 밀도를 계산하여 이상탐지에 활용하는 기법
- LOF의 이상치(Outlier) 정의: 이웃점들 대비 저밀도(low density)에 해당하는 관측치

아래 상황에서 A_1 , A_2 중에 누가 이상치에 가까운가?



LOF 모델 구축 절차

- Local Outlier Factors (LOF)

- ✓ Definition 1: k-distance of an object p

- For any positive integer k, the k-distance of object p, denoted as $k\text{-distance}(p)$, is defined as the distance $d(p, o)$ between p and an object o in D such that
- for at least k objects o' in $D \setminus \{p\}$ it holds that $d(p, o') \leq d(p, o)$
- for at most k-1 objects o' in $D \setminus \{p\}$ it holds that $d(p, o') < d(p, o)$
- Simply it is the distance to the k-th nearest neighbor considering ties.

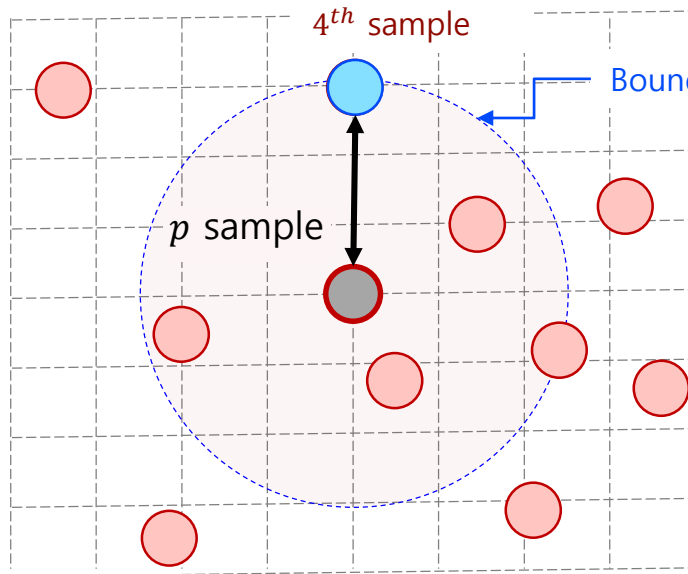
LOF 모델 구축 절차

- LOF 구축 절차:

1. $k - distance(p)$ 계산 ($k=4$ 로 사용자가 지정)

1-A. p 관측치와 네번째로 가까운 관측치를 찾아라.

1-B. p 관측치와 네번째로 가까운 관측치 간의 거리를 p 에 대한 k -distance로 저장



p 에 대한 k -distance 값은?

=

LOF 모델 구축 절차

- Local Outlier Factors (LOF)

✓ Definition 2: k-distance neighborhood of an object p

- Given the k-distance of p, the k-distance neighborhood of p contains every object whose distance from p is not greater than the k-distance

$$N_k(p) = \{q \in D \setminus \{p\} \mid d(p, q) \leq k - \text{distance}(p)\}$$

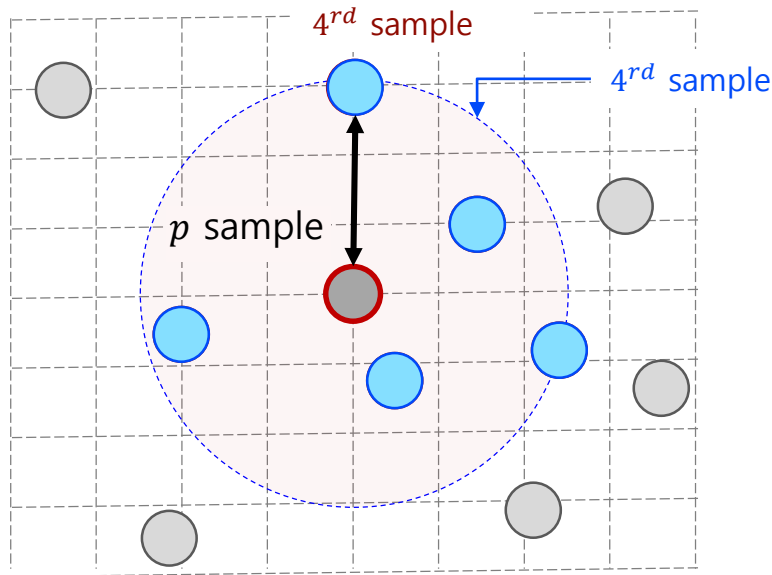
LOF 모델 구축 절차

LOF 구축 절차:

2. k -distance 내 이웃의 수 측정, $N_4(p)$ (네번째 이웃 거리내 모든 이웃의 수)

2-A. k -distance 이내 가까운 이웃 관측치들의 수를 더하기 (단, 동률도 포함)

$k + ?(k -)$



• p 의 k -distance 이내 이웃은?

=

• k -distance 이내 이웃의 수?

=

LOF 모델 구축 절차

LOF 구축 절차:

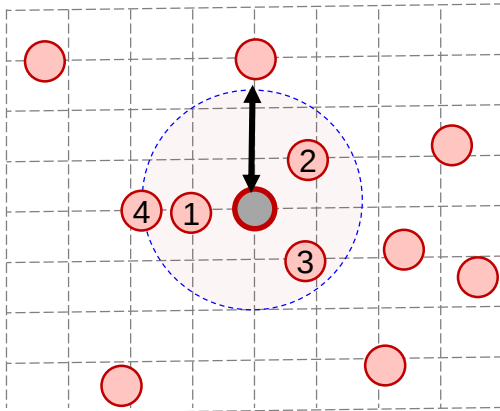
2. k -distance 내 이웃의 수 측정, $N_4(p)$

2-A. k -distance 이내 가까운 이웃 관측치들의 수를 더하기 (단, 동물도 포함)

(예제)

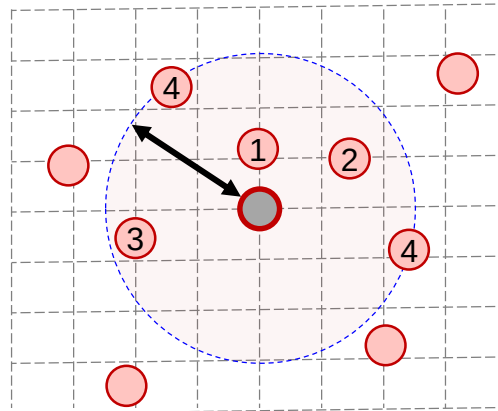
$k = 4$

k -distance = 2
 $N_4(p)$ = 4



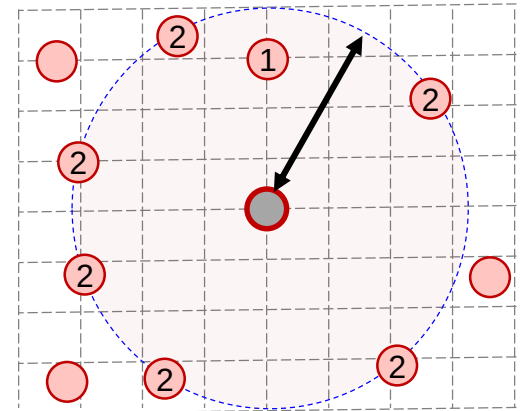
$k = 5$

k -distance = 3
 $N_4(p)$ = 5



$k = 4$

k -distance = 4
 $N_4(p)$ = 7



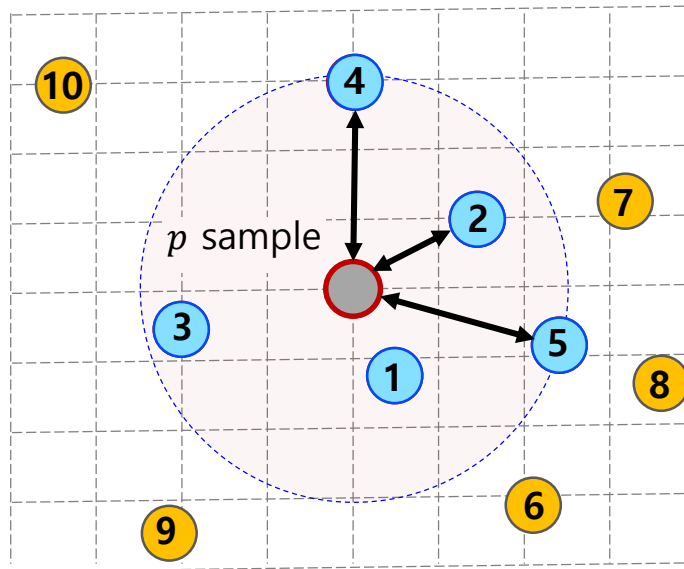
LOF 모델 구축 절차

LOF 구축 절차:

3. *Reachability* – $distance_k(p, o)$ 계산, $R_4(p, o)$ (각 이웃 간의 인접성 거리 측정)

3-A. 이웃점에 대한 인접성 거리 계산식.

$$R_k(p, o) = \text{Max}\{k - \text{distance}(p), \text{distance}(p, o)\}$$



Sample	$R_4(p)$
	$k - \text{distance}$
	$\text{distance}(p, o)$

LOF 모델 구축 절차

LOF 구축 절차:

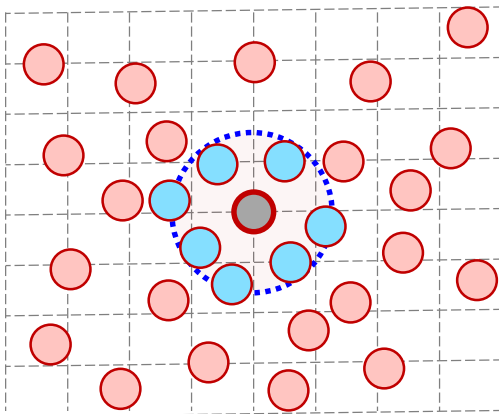
4. *Local reachability density_k* 계산, $lrd_4(p)$ (로컬 밀도거리 측정)

4-A. 관측치에 대한 상대적 밀도거리 계산.

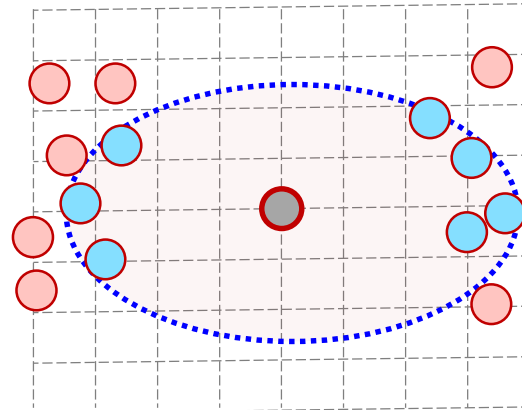
$$lrd_k(p) = \frac{|N_k(p)|}{\sum_{o \in N_k(p)} R_k(o, p)}$$

reachability distance lrd, $lrd = 1 / rd$

경우 1: **고밀도** 지역 위치 관측치 p



경우 2: **저밀도** 지역 위치 관측치 p



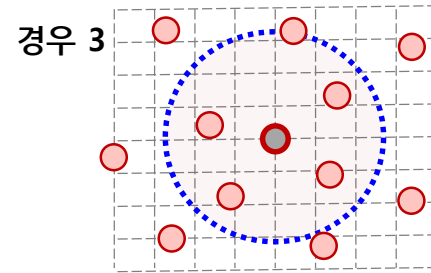
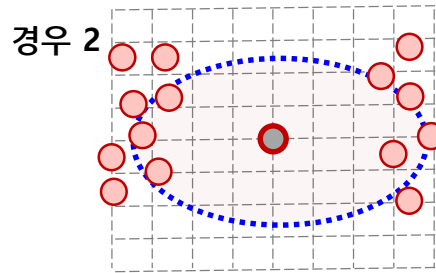
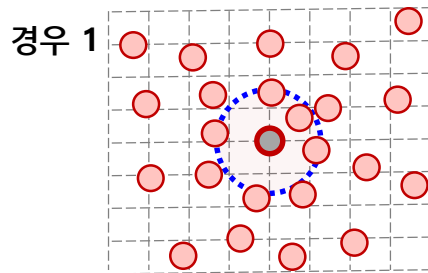
LOF 모델 구축 절차

LOF 구축 절차:

5. *Local outlier factor*_k 계산, $lof_k(p)$ (로컬 밀도거리 측정)

5-A. 관측치에 대한 상대적 밀도거리 계산.

$$lof_k(p) = \frac{\frac{1}{lrd_k(p)} \sum_{o \in N_k(p)} lrd_k(o)}{|N_k(p)|}$$



$lrd_k(o)$

$lrd_k(p)$

$lof_k(p)$

경우 1: 이웃 고밀도, 관측치 p

Large

Large

Small

경우 2: 이웃 고밀도, 관측치 p

Large

Small

Large

경우 3: 이웃 저밀도, 관측치 p

Small

Small

Small

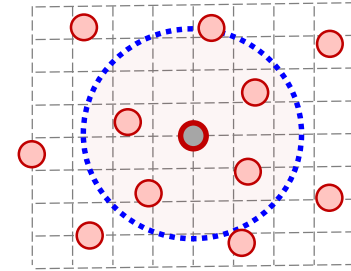
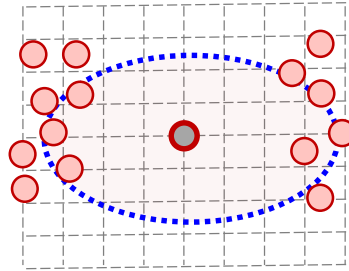
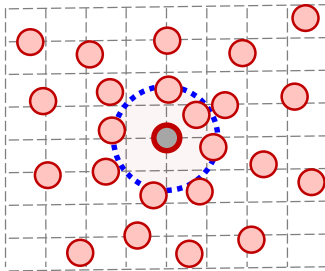
LOF 모델 구축 절차

LOF 구축 절차:

5. Local outlier factor_k 계산, $lof_4(p)$ (로컬 밀도거리 측정)

5-A. 관측치에 대한 상대적 밀도거리 계산.

$$lof_k(p) = \frac{\frac{1}{lrd_4(p)} \sum_{o \in N_k(p)} lrd_k(o)}{|N_k(p)|}$$



해석>

경우 1: 이웃 고밀도, 관측치 p

경우 2: 이웃 고밀도, 관측치 p

경우 3: 이웃 저밀도, 관측치 p

해석

이웃점도 고밀도이고, 자신도 고밀도 이므로, 이상치 일 가능성 적음

이웃점은 고밀도이지만, 자신은 저밀도 이므로, 이상치 일 가능성 높음

이웃점도 저밀도이고, 자신도 저밀도 이므로, 이상치 일 가능성 적음

LOF 모델의 이상탐지 예측

LOF 모델 구축 시 고려사항

✓ **k**를 어떻게 정할 것인가?

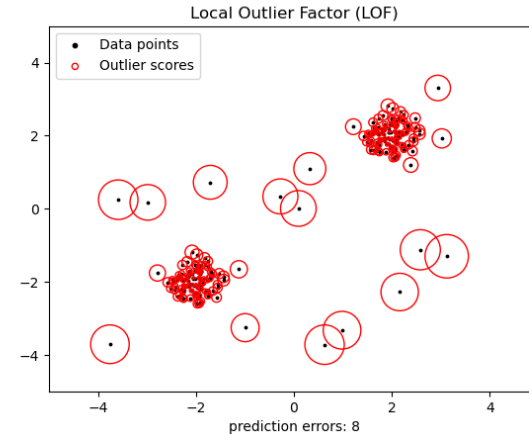
- Reference set 참조
- 가시화 후 사용자 지정
- 관측치 별 LOF 합당성 평가

✓ 얼마만큼의 LOF 점수가 나쁜 것인가?

- 유의수준 $\alpha = 0.05$ 설정
- 해당 **z score** 넘을 경우,

이상치로 간주

전체 학습 데이터
(n=100)



ID	...	LOF	
	...	53.123	5%
	...	42.211	
	...	23.333	
	...	21.002	
	...	11.005	
	...	10.859	95%
	z score	...	
	...	...	
	...	...	
	...	...	
	...	...	
	...	...	
	...	...	
	...	...	
	...	0.025	
	...	0.017	

학습 데이터 내
정상치로 간주

LOF 모델 결과 가시화 및 해석

LOF 모델 가시화

- ✓ 각 관측치의 주변 빨간색 원의 크기가 *lof score*를 의미 (원이 클수록 이상치에 가까움)

